Fakta o klimatu

Atlas of Climate Change

A Visual Guide to the Science and Data

Atlas of Climate Change: A Visual Guide to the Science and Data
Second edition, revised and extended

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Fakta o klimatu (Facts on Climate) is an interdisciplinary team of analysts based in Czechia specialising in climate change and the transition to a low-carbon economy. Through scientific data visualisations and original analytical work, the team draws attention to critical issues while providing clear, accessible information to journalists, politicians, business leaders, students, and the general public.

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Preface

Most people know that our planet is warming. The vast majority also understand that this shift is anthropogenic – a consequence of burning fossil fuels, deforestation, and other human activities. Yet, for many, the understanding stops there. The mechanics of the greenhouse effect, the carbon budgets, the scale of sea level rise, and the potential danger of tipping points remain obscured behind technical jargon or overly dense academic reports. This knowledge gap is frequently filled by noise: the topic has become so emotionally and politically charged that people are often left choosing between the paralysis of fear and the comfort of denial.

The **Atlas of Climate Change** aims to provide a data anchor, offering a sense of orientation within the vast topic of climate science. Just as a traditional map allows travellers to grasp the scale of a journey, the graphs and charts in this Atlas help readers to perceive proportions – distinguishing between natural variability, long-term trends, and large systemic changes. By grounding the debate in clear, visual insights, we hope to provide a reliable foundation for a more informed conversation.

As a team of authors, our primary challenge was not finding data, but curating it. We aimed to distill thousands of pages of research into a focused summary that captures what truly matters. Our work was guided by three deliberate editorial choices:

1. **Grounding in science**. Every key claim is linked to its primary source – often an IPCC report, a NASA dataset, or peer-reviewed literature. While academic research usually focuses on the "unknowns" at the frontiers of science, we have prioritised explaining the established climate physics most vital for public understanding.

2. **Balancing accuracy and clarity**. The volume of data regarding our changing climate is enormous. To help readers build a functional understanding, we needed to choose how to simplify the complexity. Our approach prioritises the overarching logical chain – mapping the journey from greenhouse gas emissions and their rising concentrations to the resulting heat buildup and its eventual impacts. By focusing on these essential links rather than every atmospheric variable, we ensure the big picture remains visible. For readers wishing to dive deeper into these topics beyond the scope of this Atlas, we encourage exploration of the source literature listed in our references section.

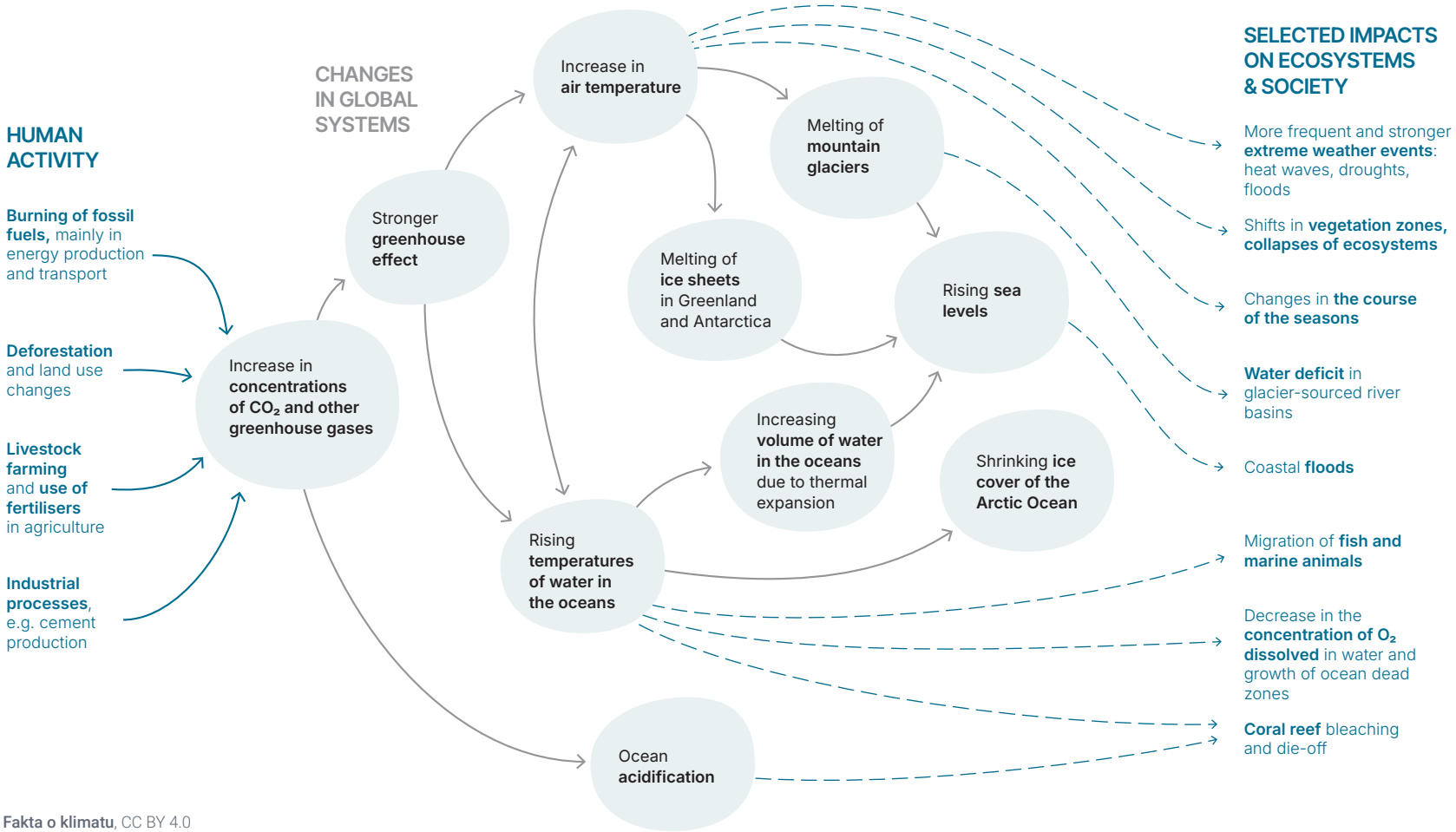
3. **A modular structure**. This Atlas is intended to be used, not just read. Every spread is designed as a self-contained overview of a specific topic, allowing the reader to start anywhere. The Atlas is built to serve multiple needs: providing a clear introduction for secondary school students, acting as a factual reference for public debate, and offering a comprehensive overview for the general reader.

Ultimately, our wish is that these pages help move the conversation from emotion toward clear understanding. We believe that clarity creates agency – and that a shared foundation of facts is the essential first step for the journey ahead.

Ondřej Přibyla & Petr Daniš

Causes and Effects of Climate Change

Climate change is much more than rising temperatures. It is a long chain of interconnected causes and effects.



Fakta o klimatu, CC BY 4.0

Climate change is not just a change in temperatures – this umbrella term covers several interrelated phenomena. Higher concentration of greenhouse gases in the atmosphere results in a long chain of interconnected impacts.

What causes climate change?

Human activity, particularly **burning fossil fuels** (coal, oil, and natural gas), leads to an **increase in the concentration of carbon dioxide (CO_2) in the atmosphere**. About 35–40 billion tons of CO_2 are produced annually in energy production, transportation, and industry, while deforestation adds another 4–5 billion tons. On average, every person on Earth accounts for about 5 tons of CO_2 per year. (See more on CO_2 concentrations and global greenhouse emissions on pp. *19* and *33*).

Other greenhouse gases include methane (CH_4), nitrous oxide (N_2O), and F-gases (fluorinated gases). Methane is produced in agriculture (livestock farming, rice cultivation, etc.) and released into the atmosphere from coal mining and oil and gas extraction. Nitrous oxide arises from the use of fertilizers and in some combustion processes. F-gases are used for refrigeration and air-conditioning. (See *Which Greenhouse Gases Matter Most? on p. 31*).

Higher concentrations of CO_2 and other greenhouse gases in the atmosphere **lead to a stronger greenhouse effect**. The natural greenhouse effect is necessary to sustain life on Earth – without it, the planet would be too cold – but a stronger greenhouse effect makes the planet warmer. (See more on *The Greenhouse Effect and Energy Imbalance on p. 21*).

What are the effects of climate change?

Atmospheric temperature has risen on average by 1.3 °C above pre-industrial levels as of 2025. However, most of the heat trapped by the stronger greenhouse effect has been absorbed by ocean water, and ocean temperatures have been rising steadily over the same period. (See more on temperature changes on pp. *13* and *15*).

The warming of the planet leads to more frequent and more severe heat waves, more intense tropical storms and hurricanes, longer periods of drought, but also to heavier rains and floods – i.e. to **more frequent extreme events**. One event often leads to another, for example, prolonged droughts make forest fires more widespread and severe. (See more on *Extreme Weather Is Growing More Severe on p. 25*)

Atmospheric CO_2 partially dissolves in the ocean, where it forms carbonic acid. This leads to a drop in pH, or ocean acidification, which is dangerous for marine life.

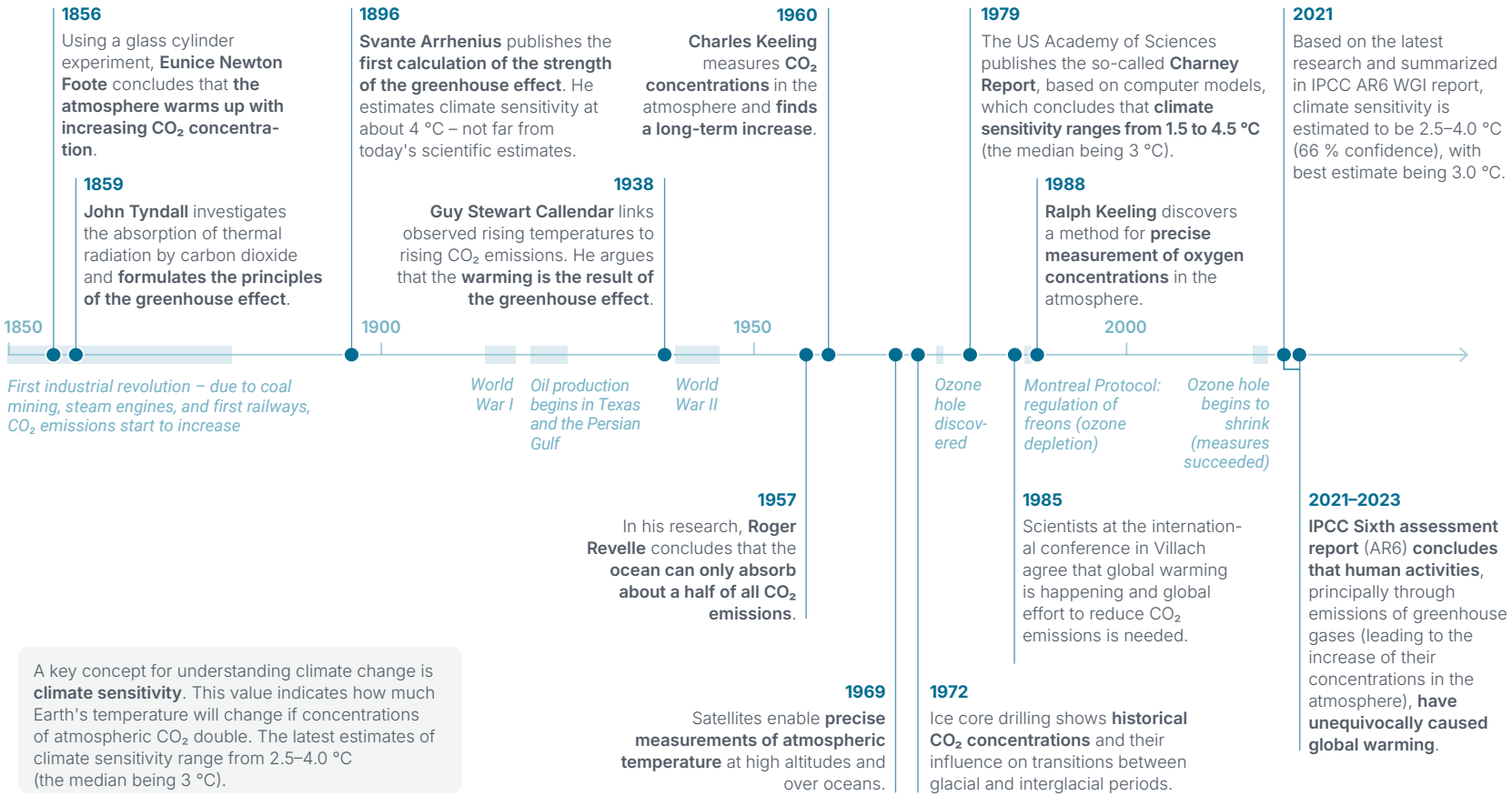
Warmer sea water temperatures are disrupting marine ecosystems and causing a significant reduction in the area and thickness of sea ice in the Arctic Ocean. Most climate models predict that the first ice-free summers in the Arctic Ocean will arrive before 2050. This sea ice melt also accelerates the warming of the Arctic region, as dark sea water absorbs much more solar radiation than white sea ice. (See more on *Accelerating Sea Level Rise on p. 23*).

The level of the world's oceans is rising at a rate of 3–4 mm per year. Roughly half of this rise is caused by the melting of ice sheets in Greenland and Antarctica, while the other half is due to the warming of seawater which increases its volume with temperature. (See more on sea level rise and climate tipping points on pp. *23, 27* and *29*).

Rising temperatures cause melting of mountain glaciers in the Alps, Himalayas, Andes, and other mountain ranges. This glacial melt will significantly impact water supplies and agriculture in the future, as rivers in many areas are fed by mountain glaciers. (See more on *Map of Climate Tipping Points on p. 27*).

Century of Climate Science

We have known about climate change caused by CO_2 emissions for more than a hundred years.



A key concept for understanding climate change is **climate sensitivity**. This value indicates how much Earth's temperature will change if concentrations of atmospheric CO_2 double. The latest estimates of climate sensitivity range from 2.5–4.0 °C (the median being 3 °C).

Fakta o klimatu, CC BY 4.0

The strength of the greenhouse effect was first estimated over a hundred years ago. More recent calculations and simulations have only made this estimate slightly more accurate: if the concentrations of atmospheric carbon dioxide double, the planet will be about 3 °C warmer.

Due to rapidly advancing science in the nineteenth century, people learned both about the composition of the atmosphere (78% nitrogen, 21% oxygen, 1% argon and small amounts of other gases including CO_2) and about CO_2 absorbing thermal radiation. With the industrial revolution well underway and more intense mining and burning of coal, some scientists began to wonder whether CO_2 concentrations had increased, and if so, whether this might affect the planet's temperature.

The first calculation of the greenhouse effect was made in 1896 by Svante Arrhenius. Beside the effects of CO_2, his calculation included the effect of water vapour, which also acts as a greenhouse gas. Its amount in the atmosphere depends on temperature – warmer air can hold more vapour, and when temperature drops, the vapour condenses and falls as precipitation. Arrhenius concluded that if CO_2 concentrations in the atmosphere doubled, the planet would be about 4 °C warmer due to the combined effect of CO_2 and the increased water vapour present in warmer air.

***Climate sensitivity** means how much warmer (or cooler) Earth's surface will get in relation to the change of atmospheric CO_2 concentrations. If the CO_2 concentrations double, it is now estimated that the planet will get 2.5–4.0 °C warmer (the best estimate is 3 °C).[1] This is roughly equivalent to saying that each additional 1,000 Gt of CO_2 in the atmosphere leads to approximately 0.45 °C of warming.[2]*

Early computer simulations of global climate began to appear in the **1970s**. Physicists who made them already knew the exact concentrations of CO_2 in the atmosphere and how quickly they were rising (see *The Changing Balance of Atmospheric CO_2 & O_2 on p. 11*). Their simulations only made Arrhenius' estimate more accurate: if the concentrations of atmospheric CO_2 double, the planet will be about 3 °C warmer. The simulations also showed that the **warming will be significantly stronger over the Arctic**.

One important milestone was the **Charney Report (1979)**, in which scientific findings on climate change were presented to the US government. The authors **warned of the expected negative impacts** of future climate change and urged action to reduce CO_2 emissions.

Major oil companies came to the same conclusion around that time. An ExxonMobil internal report from 1982 describes in detail the relationship between Earth's temperature and the concentration of atmospheric CO_2 – with the same estimated climate sensitivity.[3]

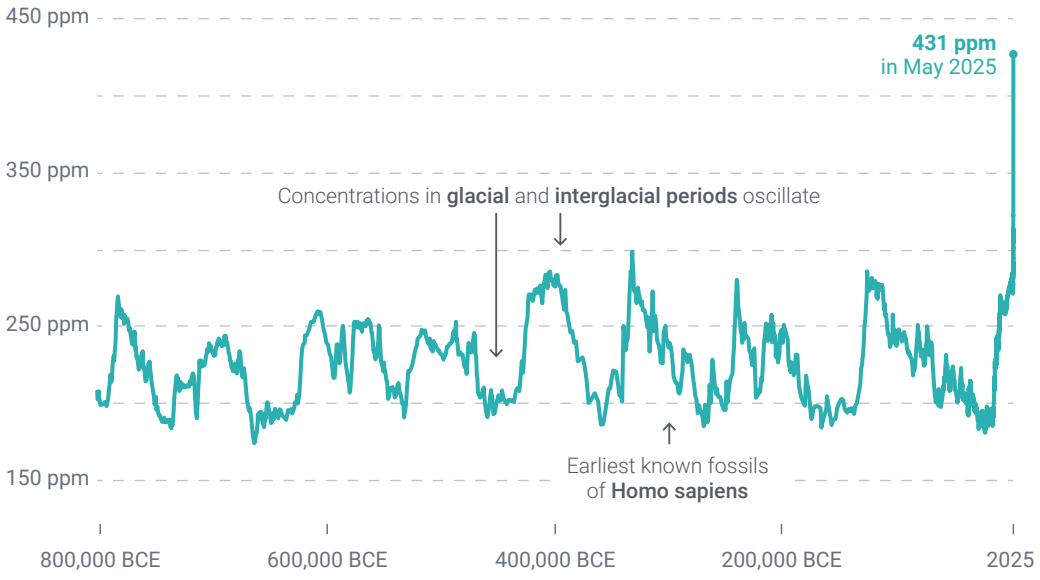
Present-day computer simulations model the temperature, humidity, and airflow across atmospheric layers, deep and surface ocean currents, and a wide range of additional factors – including geography, vegetation, clouds, sea ice, glacier trends, and many other variables. Simulations of past climate now closely match the historical record, especially the overall picture of how climate has changed over time. These **advanced simulations have further reduced uncertainties** in the original estimates **of climate sensitivity.**

When we speak of global warming, it is the average of temperatures from different areas around the world (see Global Map of Temperature Changes on p. 15).

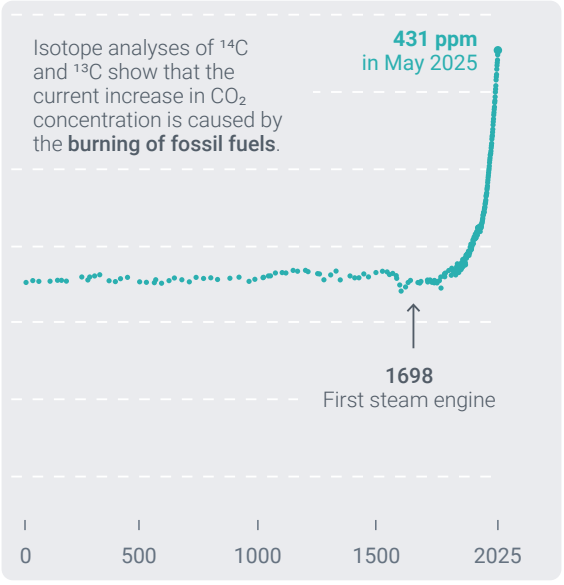
CO_2 Concentrations at an 800,000-Year High

The concentration of atmospheric CO_2 today is the highest since the dawn of humanity.

CO_2 CONCENTRATIONS OVER THE **LAST 800,000 YEARS**



ZOOM IN: **LAST 2,025 YEARS**



The unit of measurement used for concentration is **ppm (parts per million)**.

A concentration of 400 ppm CO_2 in the atmosphere means that there are 400 molecules of CO_2 in one million molecules of air.

The CO_2 concentration values come from the EPICA **analysis of samples from deep ice core drilling** in Antarctica and from **direct measurements** at Mauna Loa, Hawaii.

Fakta o klimatu, CC BY 4.0

Data source: NOAA, Antarctic Ice Cores Revised 800KYr CO2 Data; NOAA, Global Monitoring Library, Mauna Loa

The last time atmospheric CO_2 concentrations were as high as today was around four million years ago. Earth was about 3 °C warmer, sea levels 20 metres higher, mastodons and three-toed horses roamed across Europe, and the earliest human ancestors in Africa were only just beginning to walk consistently upright.

The graph on the left shows a trend in CO_2 concentrations over the last 800,000 years.[4] The data reveal how these **concentrations oscillated in glacial and interglacial periods** – from 170 ppm to 280 ppm.

Zooming in, the graph on the right has a different time scale and shows in detail a trend in CO_2 concentrations over the last 2,000 years. **The concentrations were around 280 ppm until the industrial revolution**; then they started to rise.

How do scientists know the composition of the atmosphere in the past?

Samples of ancient air can be found **as bubbles trapped in the ice sheets** of Greenland or Antarctica. If you take a sample cut from an ice core drilled deep into an ice sheet and melt it in a laboratory, you can analyse the atmospheric composition at the time the ice formed. The age of the sample depends on how deep you drill. The Antarctic ice sheet is so thick that cores drilled nearly three kilometres down contain air bubbles around 800,000 years old.

Could present-day climate change be a part of a natural cycle?

We live in an interglacial period and if the cycle continued naturally, Earth would be slowly cooling now – CO_2 concentrations would be slowly declining and another glacial period would start within tens of thousands of years.

The natural cycling between glacial and interglacial periods is driven by variations in Earth's orbital shape as well as the tilt and orientation of its axis, collectively known as Milankovitch cycles. This process triggers a complex system of interrelated changes in which CO_2 as a greenhouse gas plays a major role. But Milankovitch cycles operate on timescales of tens to hundreds of thousands of years – which is why they cannot explain the rapid rise of CO_2 seen in the last 200 years.

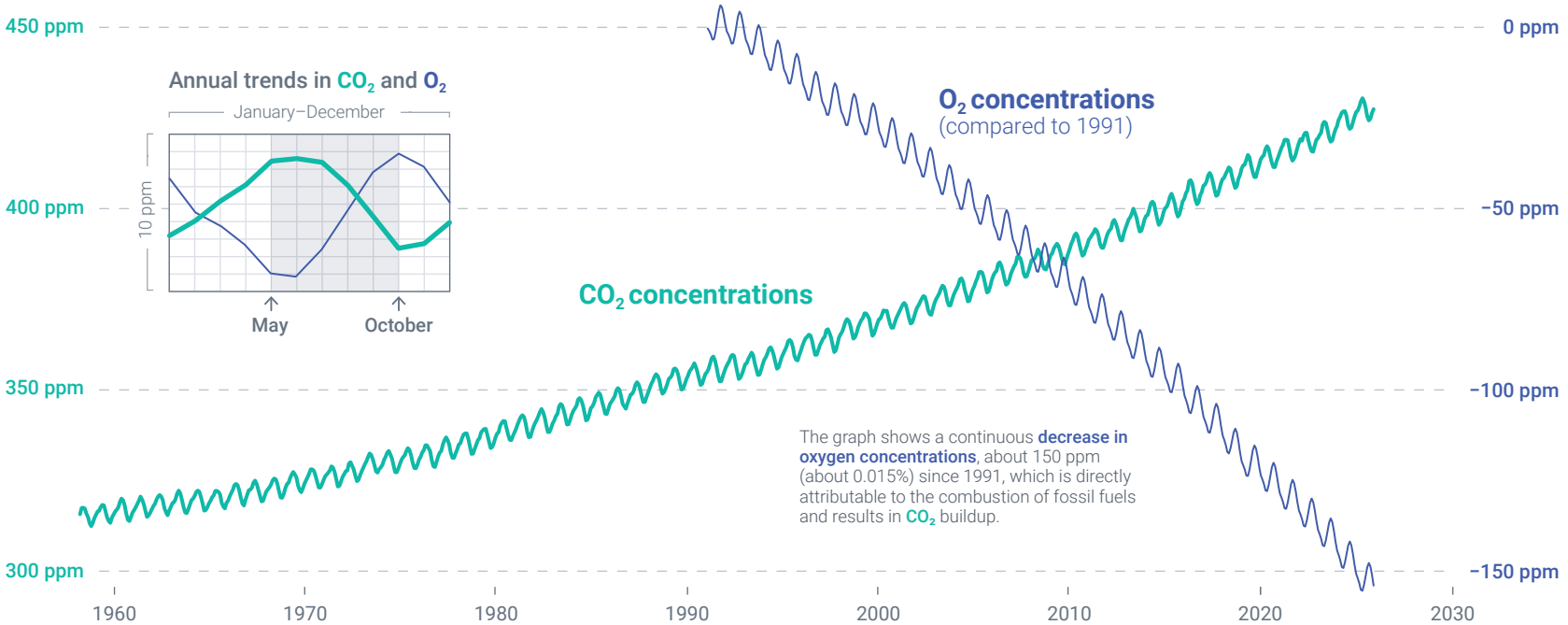
The climate change happening now is unrelated to the natural variation of glacial and interglacial periods – current CO_2 concentrations are significantly above the range typical for this variation, and the amount of solar radiation determined by Milankovitch cycles shows no unusual trends for our times.

Moreover, there is ample evidence that the rise in CO_2 concentrations is caused by fossil fuel combustion:

- → The rise of CO_2 concentrations corresponds to the amount of fossil fuels burned each year (when some absorption of CO_2 by the ocean is factored).
- → Carbon atoms in CO_2 are not all the same – they exist as different isotopes. Different sources of CO_2 have unique isotopic fingerprints. Because fossil fuels are millions of years old, they contain virtually no ^{14}C, which decays over time. Isotope analyses of atmospheric CO_2 show declining concentrations of ^{14}C, meaning that an increasing proportion of atmospheric CO_2 comes from fossil fuel emissions.[5]
- → The rise of CO_2 concentrations is paralleled by the decline of O_2 concentrations, which corresponds to the O_2 loss by burning fossil fuels (see *The Changing Balance of Atmospheric CO_2 & O_2 on p. 11*).

The Changing Balance of Atmospheric CO_2 & O_2

Changes in carbon dioxide (CO_2) and oxygen (O_2) concentrations in time show annual cycles of respiration and photosynthesis on Earth as well as long-term effects of fossil fuel combustion.



Concentrations of CO_2 are measured in units of ppm (parts per million). A concentration of 400 ppm CO_2 in the atmosphere means that there are 400 molecules of CO_2 in one million molecules of air, which corresponds to a concentration of 0.04% CO_2 in the atmosphere.

Air contains 78.1% nitrogen, **20.9 % oxygen**, 0.9 % argon and approximately **0.04 % CO_2**.

Fakta o klimatu, CC BY 4.0

Data source: NOAA, Global Monitoring Library, Mauna Loa; Scripps O2 Program, Mauna Loa Observatory

A long-term trend in increasing concentrations of carbon dioxide (CO_2) and declining concentrations of oxygen (O_2) shows that humans have been causing changes in the atmosphere composition by their activity – primarily by burning fossil fuels.

CO_2 concentrations[6] oscillate during the year, but **are increasing in the long term: about 26 ppm just in the last decade** (2015–2025). Concentrations were around 315 ppm in 1960, and reached around 425 ppm in 2025 – which means they have risen by 35% in only 65 years.

The graph also shows a trend in **O_2 concentrations**[7]: how much it changed each year compared to the reference year (1991). **O_2 concentrations** also oscillate during the year but **are declining in the long term: around 55 ppm just in the last decade**.

How are CO_2 and O_2 concentrations measured?

The method for exact determination of CO_2 concentrations (to 0.1 ppm, i.e. 0.00001%) was developed by **Charles Keeling** in 1952. He was surprised by the results of his measurements at first as the concentrations appeared to be changing randomly, depending on the direction of the wind. Then he realized that his results in San Francisco were influenced by nearby forests (photosynthesis) and local factories (combustion) and that concentrations must be measured at a place which is far enough from such factors. So he moved to the central Pacific: to Mauna Loa in Hawaii. It was not until he got to this remote location that his measurements started to make sense – concentrations proved stable in Hawaii. Keeling noticed that **concentrations oscillate during the year – falling from May to October and rising in the remaining part of the year**. He was observing Earth's "breathing".

During photosynthesis, plants capture CO_2 from the air and produce O_2. Conversely, they use O_2 and release CO_2 during respiration.

$CO_2 + H_2O \rightleftarrows O_2 + carbohydrates$

Most of the world's forests are located in the Northern Hemisphere. In summer, when deciduous trees are in full leaf, photosynthesis prevails; plants capture CO_2 and use the carbon to build their trunks and foliage. In winter, photosynthesis slows down dramatically and plants begin to release CO_2 through respiration, as they burn energy to maintain their cells, even in a dormant state. Meanwhile, soil microbes decompose fallen leaves, further releasing CO_2 back into the air.

Keeling observed this oscillation between summer and winter but also noticed **a long-term increase in CO_2 concentrations** which he believed was **caused by the burning of coal, oil, and natural gas**.

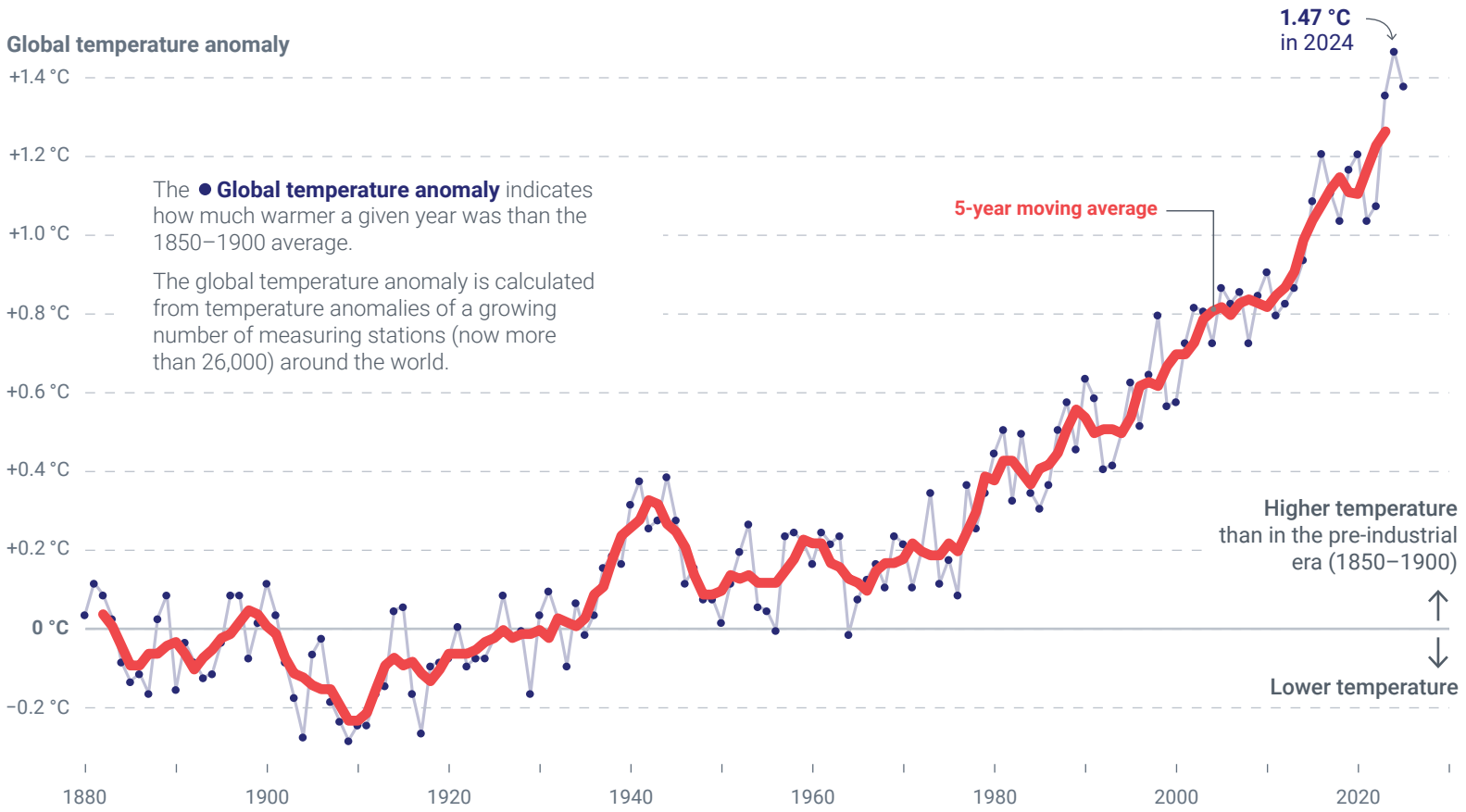
During combustion, O_2 is consumed and CO_2 released. When coal is burned, the chemical reaction is simple: $C + O_2 \rightarrow CO_2$.

When natural gas is burned, even more O_2 is consumed as water vapour is produced as well: $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$

The fact that rising CO_2 concentrations are caused by combustion was proven by Keeling's son Ralph, who found a precise method to measure concentrations of oxygen in 1988. His results show an **unnatural long-term decline in O_2 concentrations**. Subsequent research, including isotopic analysis, confirmed that the rise in atmospheric CO_2 is attributable to fossil fuel combustion.[8]

How Much Is the Planet Warming?

The world is now approximately 1.3 °C warmer than in 1850–1900. Over the last thirty years, the warming rate has been about 0.25 °C per decade.



Fakta o klimatu, CC BY 4.0

Data source: NASA GISTEMP v4

The data from scientific measurements show that temperatures keep rising with increasing CO_2 concentrations, in line with the predictions.

The graph shows the trend in **global temperature anomaly** (deviation from normal) **over the past 145 years**.[9] Ever since around 1910, the world has been warming. It does not mean that every year is warmer than the year before – there is a natural variation, but the trend is clear. As of 2025 the **top 10 hottest years on record were all the last 10 years** (2016–2025).

In 2024, global warming has reached 1.47 °C relative to the pre-industrial era, according to NASA, and has even surpassed the 1.5 °C threshold according to other temperature datasets.[10] However, 2024 is just a single year. Taken as a five-year average, **the world is currently** (as of 2026) **about 1.3 °C warmer** than in the pre-industrial era – in line with climate model predictions.

What is a reference period?

Whenever we say how much the global temperature has increased, we need to be clear since when (i.e. what the reference period for that warming is). For example, 2024 was 1.47 °C warmer than the average temperature in 1850–1900, but only 0.78 °C warmer than the average temperature in 1981–2010 (because the later period was on average already 0.69 °C warmer than the former).

Climate scientists often use 1850–1900 as a reference period (also called "the pre-industrial era"). This term is not strictly accurate, because the industrial revolution had already been underway at that time. But the concentrations of CO_2 in the atmosphere were around 280–300 ppm back then, and Earth's temperature was not yet affected by the strengthening greenhouse effect very much.

Any warming mentioned here is measured against this pre-industrial baseline, which is worth keeping in mind when comparing figures from other sources.

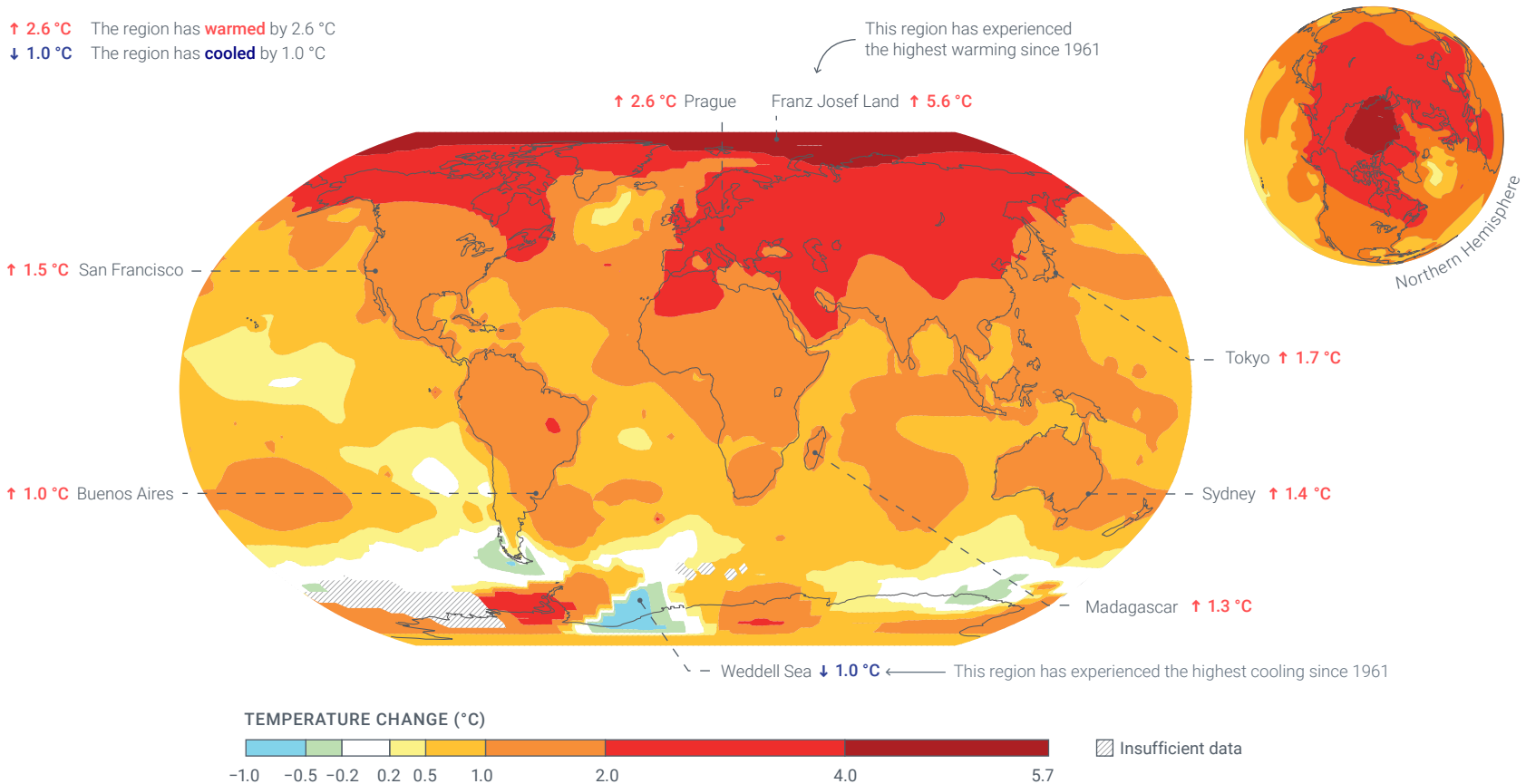
What is a temperature anomaly?

Because the Earth's surface is varied, calculating a mean temperature raises both methodological and practical difficulties. Therefore, on a global scale it is more useful to talk about a temperature anomaly. Nevertheless, mean temperatures can also be useful – e.g. for smaller entities such as cities or small countries.

A global temperature anomaly for any given year says how much warmer the world is in that year when compared to the average temperature in the reference period. The global temperature anomaly in 2024 was 1.47 °C – meaning that year was globally (on average) 1.47 °C warmer than the global average of 1850–1900. This average masks large variations between warming in different regions of the world. All continents were even warmer than global average warming of 1.47 °C in 2024 (with Europe warming at roughly twice the global average rate), as warming above land (and in the Northern Hemisphere) is stronger than warming above oceans (see *Global Map of Temperature Changes on p. 15*).

Global Map of Temperature Changes

Between 1961 and 2025, warming has been uneven across the globe. Continents have warmed faster than oceans, and the Northern Hemisphere faster than the Southern Hemisphere.



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Data source: NASA GISTEMP v4

Changes over the last six decades have shown that the warming rate varies across different regions, which is consistent with predictions from computer simulations.

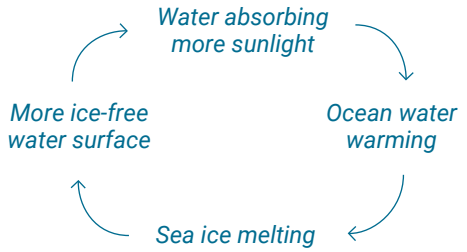
The map shows in detail how much temperatures changed between 1961 and 2025 in different parts of the world.[11] Landmasses are warming faster than oceans. The warming is the fastest over the Arctic – the climate above some Arctic islands is now more than 5 °C warmer than it was six decades ago. **Europe**, with its landmass relatively close to the Arctic, **is the fastest warming continent.**

Nonetheless, there are still regions which are not warming at all or even getting slightly cooler.

Why is the North warming so fast?

The amplification of warming over the Arctic Ocean is in large part related to the different physical properties of ice and water. While dark ocean water absorbs roughly 90–95% of incoming radiation, and reflects just 5–10%, sea ice reflects up to 90% of it – this **degree of reflectance** is called **albedo**.

Due to increasing temperatures over the Arctic Ocean, sea ice is melting. As a result, more water surface is ice-free and more solar radiation is being absorbed, which leads to further warming of the ocean, further temperature increase, more ice melting, and so on.



Why is the warming not amplified in the South?

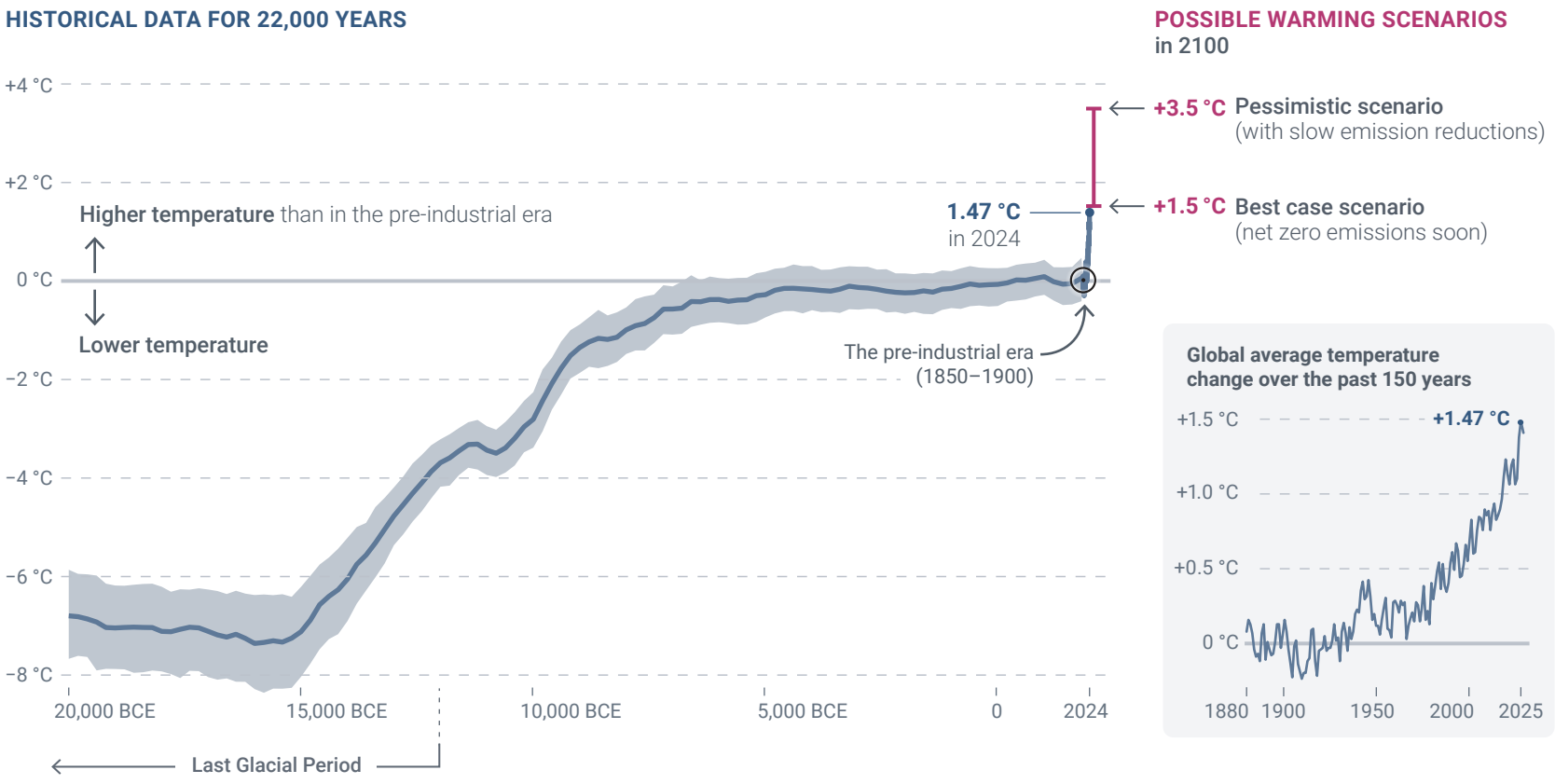
The feedback loop of sea ice melting and ocean warming is much less common in the Southern Hemisphere, as the ice in the Antarctic is mainly on land and up to several kilometres thick. When the surface melts, it just uncovers deeper layers of ice, but the albedo remains unchanged.

However, the reasons why some areas around the South Pole are experiencing less warming or even slight cooling are much more complex. E.g., stratospheric ozone acts as a greenhouse gas, so its depletion over Antarctica has had a local cooling effect in the upper atmosphere. Strong westerly winds and ocean currents (notably, the Antarctic Circumpolar Current) circle the continent as a heat barrier. And even the melting of the ice sheet (the result of global warming) adds so much fresh cold water to the sea surface that it leads to its cooling.[12]

Even so, despite such regional cooling trends, the Antarctic continent as a whole is **losing ice mass at an accelerating rate**, driven primarily by warming ocean waters melting its ice shelves from below (see more on rising sea levels and tipping points of ice sheets on pp. *23* and *27*).

Warming from the Last Glacial Period to the Present

During the transition from the Last Glacial Period, the planet warmed at a rate of about 1 °C per millennium. Currently, it is warming more than ten times faster.



Fakta o klimatu, CC BY 4.0

Data source: Osman et al. (2021); NASA GISTEMP v4; IPCC, AR6 WGI

Some argue that the climate has always been changing. The climate has indeed always changed – but the rate of current warming is unprecedented in the context of human civilization. It is the speed of change that matters for ecosystems and societies.

In the long geological timescale of Earth's history, it is possible to find periods in which the climate was much colder than today (glacial periods of ice ages) as well as much warmer (e.g. the age of dinosaurs). However, climate changes were always relatively slow.

From this historical perspective, **the temperature change in the last hundred years** shown in the graph **is huge and abrupt – such a trend is not natural for the world's climate**.

Calculations and computer simulations also show that at the end of this century the planet will be about 1.5–3.5 °C warmer than in preindustrial era, depending on how quickly greenhouse gas emissions are reduced[13] (see more on *Projections of Warming in 2100 on p. 37*).

What does the temperature change mean for humans and for the living world?

During the Last Glacial Maximum (peaking between 26,000 and 20,000 years ago), Canada and northern Europe were covered by vast ice sheets, and cold tundra stretched all the way down to the Mediterranean coast. Ice sheets locked up huge quantities of water so global ocean levels were about 120 meters lower than today.

In the following ten thousand years, temperatures rose slowly and the world became 7 °C warmer.[14] Glaciers melted away and the natural world adapted to this change – including humans who could now settle in previously uninhabitable northern areas. This warming period was followed by a long era in which the annual global temperature did not change very much, which was favourable for civilization growth.

Current warming,[15] however, is at least 10 times faster and **does not give ecosystems and species sufficient time to adapt**. This abrupt warming may have far-reaching consequences for life on the planet as well as for human civilization (see more on climate tipping points on pp. *27* and *29*).

How do scientists measure past temperatures?

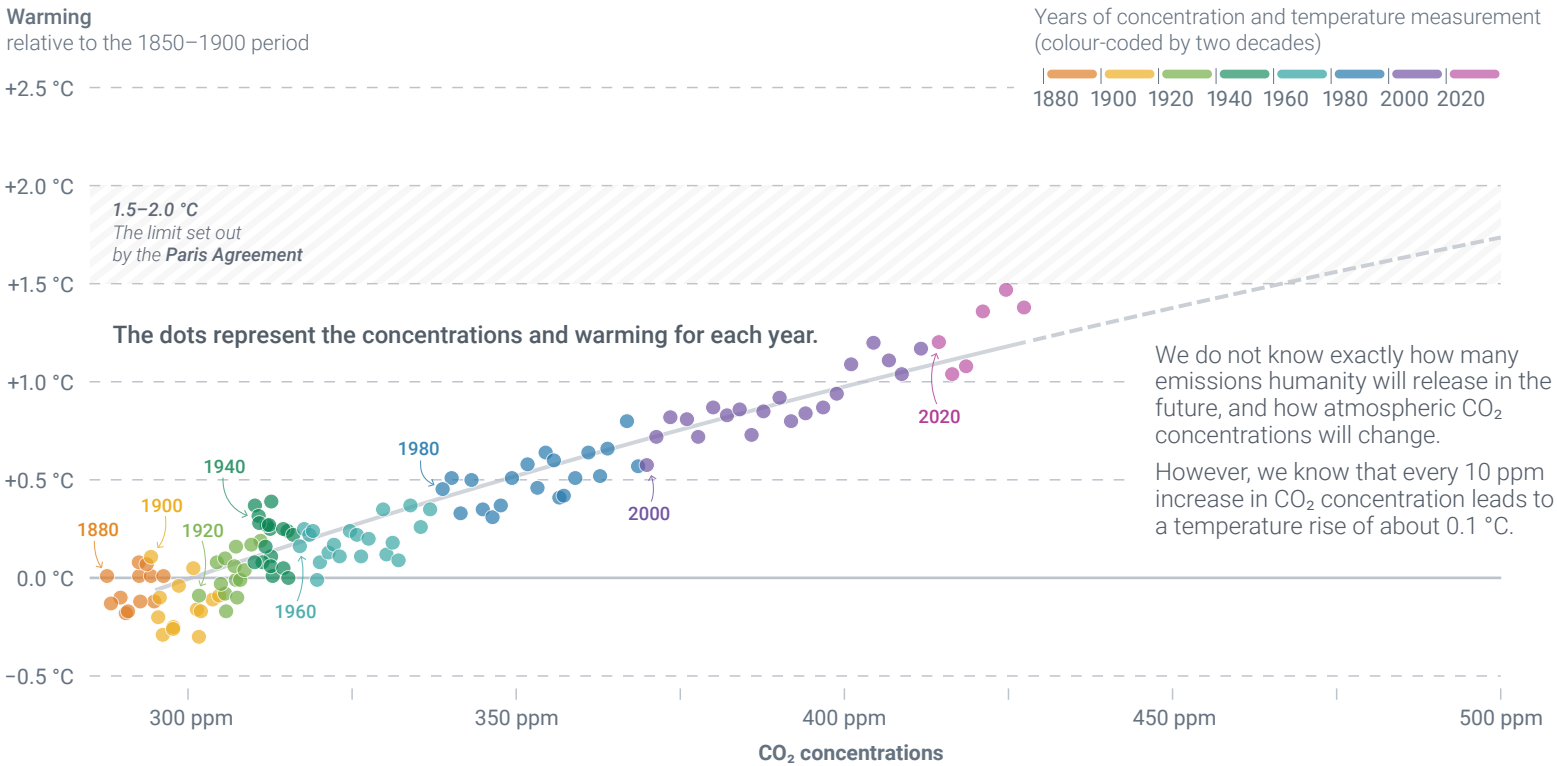
Air temperatures have been reliably measured since the 18th century. Temperatures in earlier or even pre-historic times can be calculated based on some natural processes affected by temperature changes. One example is dendrochronology – the dating of tree rings. When a piece of wood is found by archaeologists as a part of their excavations, the growth rings allow scientists to estimate temperatures in the period when the tree was growing.

Another method studies pollen grains found at bottoms of lakes – this pollen provides useful information about plants growing by the lake in the past and researchers can thus estimate the temperatures in those times.[16]

Key data about past temperatures comes from the **isotope analysis of marine sediments**. In water (H_2O), oxygen exists as different isotopes (^{16}O and ^{18}O), the ratio of which changes depending on the planet's climate. By analysing the calcium carbonate shells ($CaCO_3$) of dead organisms from the seafloor, scientists can determine the composition of ancient seawater and from that, reconstruct temperatures with reasonable accuracy even in the distant past.

The Direct Link Between CO_2 and Temperature

The higher the CO_2 concentration in the atmosphere, the higher the Earth's temperature. The levels of atmospheric CO_2 depend on the amount of emissions produced by humankind.



CO_2 concentration is measured in ppm (parts per million). The CO_2 concentration of 400 ppm means that one million of air molecules contains 400 molecules of CO_2. Carbon dioxide (CO_2) contributes to global warming more than any other greenhouse gas.

Fakta o klimatu, CC BY 4.0

Data source: NOAA, Global Monitoring Library, Mauna Loa; NASA GISTEMP v4

The graph reveals a clear relationship: the higher the CO_2 concentration, the higher the global temperature. To stop global warming, the world needs to stop CO_2 emissions.

The points in the graph show individual years in the period of 1884–2024. The location of each point corresponds to the CO_2 concentration values in Mauna Loa in that year (on the horizontal axis)[17] and the global temperature anomaly values for that year (on the vertical axis).[18]

The graph indicates that within the range of concentrations shown, the relationship is approximately linear – **each 10 ppm increase in CO_2 concentration leads to a temperature increase of about 0.1 °C** – though over larger ranges this proportionality becomes less precise.

The dots in the graph are colour-coded and grouped (each period of 20 years has its own colour). It is clear that the increase in CO_2 concentrations has been accelerating in recent years, corresponding to increasing annual CO_2 emissions.

The right part of the graph shows the expected global warming for higher CO_2 concentrations in the future, provided CO_2 emissions continue.

How are other greenhouse gases and aerosols affecting the warming beyond CO_2?

Carbon dioxide is responsible for over half of observed global warming. The remaining part is due to **other greenhouse gases**, mainly methane, nitrous oxide, and F-gases: their concentrations in the atmosphere are also increasing.[19] (See more on all main greenhouse gases on p. *31*).

Along with greenhouse gases, humankind also emits **aerosols**, which have **a cooling effect** on the planet – they reflect solar radiation and participate in the formation of clouds.

The global warming shown in the graph on the vertical axis includes all of these phenomena, but only CO_2 concentrations are plotted on the horizontal axis. Thus, the claim of proportionality between increasing concentration of CO_2 and warming is somewhat simplified. While **CO_2 concentration is the dominant factor**, it is not the only one.

What happens if emissions stop?

As simple as this question seems, the answer is complex and depends entirely on which emissions are stopped.

If only **CO_2 emissions** were to **stop** – the dominant warming factor – global **temperatures** would **stabilise within a few years**.[20]

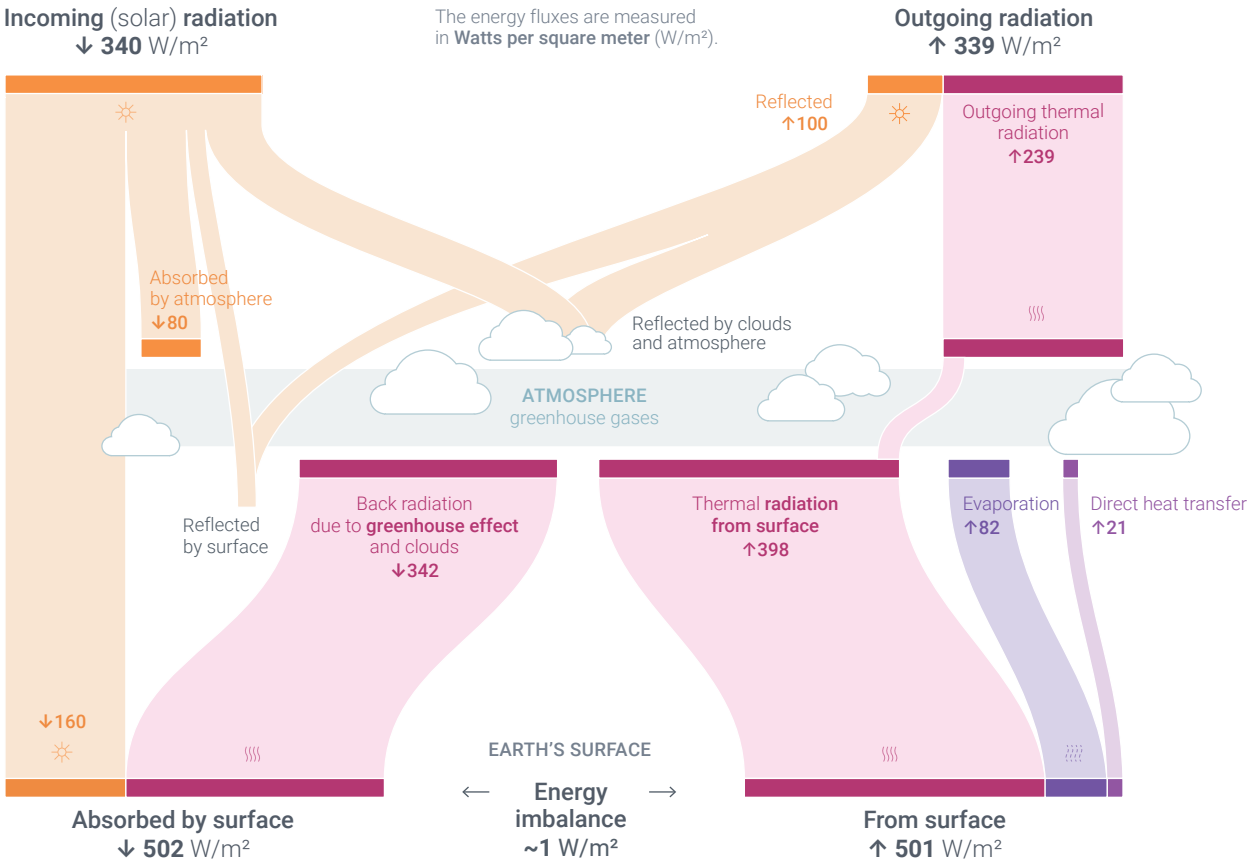
However, in the real world, **CO_2 emissions are closely tied to aerosols**. Most cooling aerosols are formed from substances such as sulphur dioxide, which are direct byproducts of burning the same fossil fuels that produce CO_2.

So if the world **stopped** both **CO_2 and related aerosol emissions** simultaneously, **temperatures** would actually **rise** – first rapidly, then more gradually – **by about 0.5 °C**. This is because the cooling "mask" of aerosols washes out of the atmosphere, while the CO_2 remains.[21]

Finally, if **all greenhouse gas and aerosol emissions stopped**, we would see an initial rapid temperature spike as aerosols disappeared, followed by a gradual cooling trend as methane and other short-lived greenhouse gases broke down. This would eventually lead to a **slight** overall **decrease in global temperature**.[22]

The Greenhouse Effect and Energy Imbalance

The Earth's climate is determined by a radiation balance between the energy received from the Sun and the energy emitted back into space. Human-induced changes to the atmosphere have now disrupted this equilibrium.



The **greenhouse effect** gives our planet a temperature hospitable to life.

Greenhouse gases allow **shortwave radiation from the Sun** to reach Earth, but they absorb **longwave thermal radiation from Earth's surface** and send a large portion of it back.

The Earth is normally in a state of radiative equilibrium. Currently, with stronger greenhouse effect, it moved to **energy imbalance**, when it **absorbs more energy (~1 W/m²)** than it radiates to space.

Fakta o klimatu, CC BY 4.0

Data source: IPCC, AR6 WGI

The Earth stays habitable because of a delicate balancing act. It sits in the cold vacuum of space, constantly receiving an enormous flow of solar energy that heats it up. To keep its temperature stable, it must release an equal amount of energy back into space. Currently, greenhouse gases emitted by humans have upset this balance, leading to an accumulation of thermal energy within the Earth system.

Without an atmosphere, the Earth would be a frozen desert with an average temperature of -18 °C. Instead, our planet averages a life-sustaining 15 °C. This 33 °C difference is **provided by the greenhouse effect**.

Greenhouse gases act like a **one-way filter**: they are transparent to incoming shortwave solar radiation but absorb the longwave thermal radiation emitted by the Earth as it passes through the atmosphere. Then they re-radiate that energy in all directions and send a substantial portion back down toward the ground – called back radiation – which warms the surface to the current level.

The most powerful greenhouse gas is **water vapour** (H_2O), responsible for over half of the effect. It acts as an amplifying feedback: as the air warms, it holds more vapour, which traps more heat in turn.

While much less abundant than water vapour, **carbon dioxide (CO_2)** plays a crucial role. Because it stays in the atmosphere for centuries, it sets the baseline temperature that allows other factors, like water vapour, to react.

How does the Earth exchange energy?

The Earth's climate is governed by the exchange of radiation between space, the atmosphere, and the planet's surface. The energy fluxes are measured in Watts per square meter (W/m^2).[23]

Incoming solar radiation: The Sun delivers approximately 340 W/m^2 of shortwave light. About 100 W/m^2 of this is immediately reflected away by clouds, atmospheric particles, and bright ice. The remaining 240 W/m^2 is absorbed by the surface (160 W/m^2) and the air (80 W/m^2).

Thermal radiation from the surface: Like all physical objects, the Earth emits thermal radiation proportional to its temperature – about 398 W/m^2.

Additional heat transfer: Energy also leaves the surface through **direct heat transfer to the air** (sensible heat, 21 W/m^2) and through **water evaporation** (82 W/m^2 – energy stored in water vapour and released higher in the atmosphere when the vapour condenses).

Back radiation: The atmosphere with greenhouse gases intercepts much of this thermal radiation and sends it back to the ground (342 W/m^2).

Outgoing thermal radiation: 239 W/m^2 escapes from the Earth to space. To maintain a stable temperature, the planet should lose an equal amount of energy as it receives from the Sun.

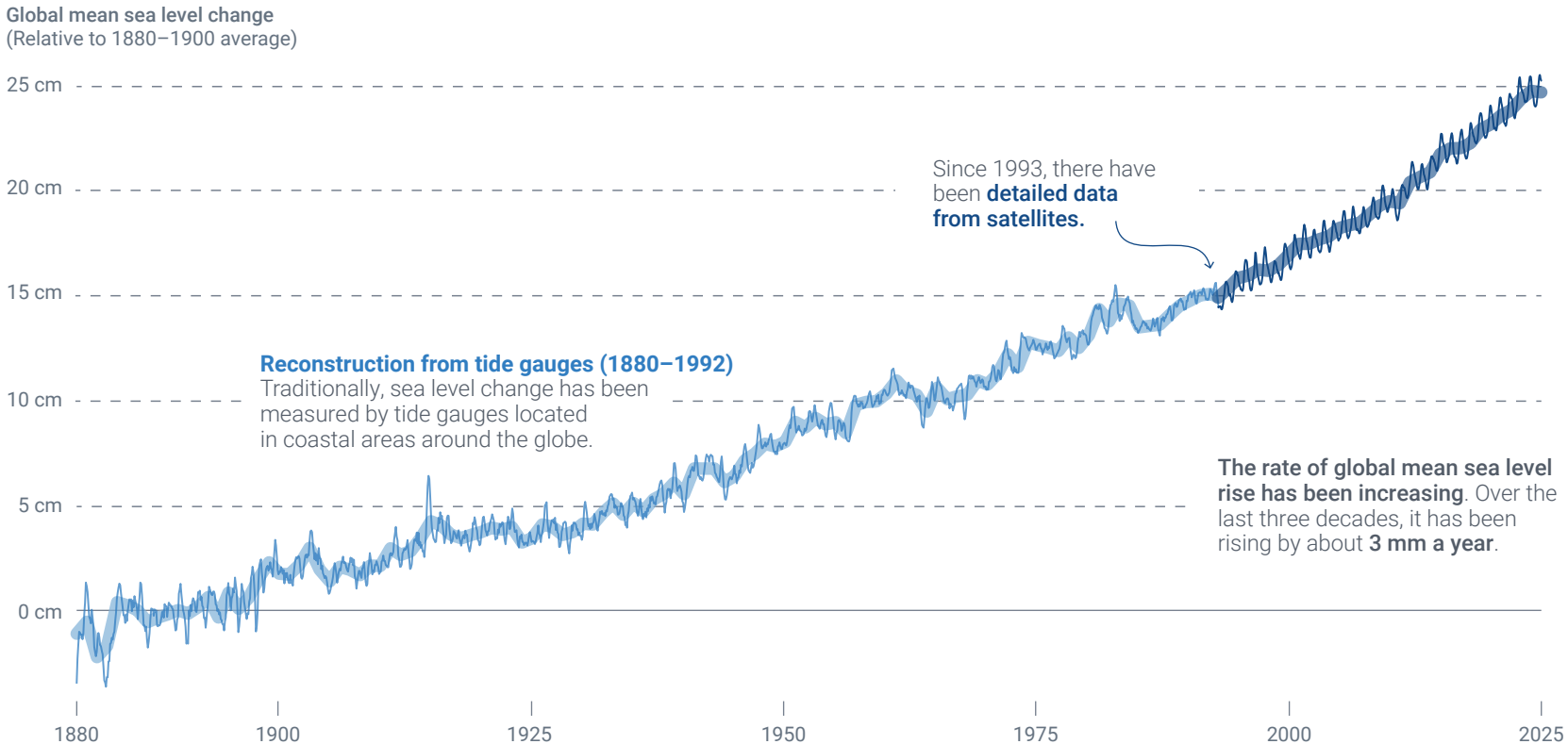
Why is there an energy imbalance now?

By emitting greenhouse gases, **humans have strengthened the greenhouse effect**. The graph shows that the radiation going **out** of the Earth system is about 1 W/m^2 lower than the radiation coming **in**, resulting in about 1 W/m^2 more energy (and growing in the last decades)[24] warming the Earth's surface and creating an energy imbalance.

A single Watt per square meter may sound negligible, but the scale of our planet turns this into a massive amount of energy – equivalent to the **energy of eight Hiroshima-style atomic bombs exploding every second**. Over 90% of this surplus heat is being absorbed by the oceans, the rest by land, by ice sheets, and (about 1%) by the atmosphere.[25] Until humanity stops emitting greenhouse gases, the planet will continue to store more heat.

Accelerating Sea Level Rise

From 1880 to 2025, global mean sea level rose by about 25 centimetres. It will continue to rise for centuries as oceans warm and ice sheets and glaciers melt.



Fakta o klimatu, CC BY 4.0

Data source: CSIRO 2017, NASA-SSH

Sea level is rising in most areas of the world and will keep rising for many more centuries due to ice loss and rising ocean temperatures caused by climate change. By 2100 some places could see around 1 meter of sea level rise, but the exact amount depends on future levels of warming.

Global mean sea level is the **average height of the entire ocean surface**. The graph shows the change in sea level since 1880 measured by tide gauges located around the coasts,[26] and by satellites over the last three decades.[27] Although there are years when global mean sea level rises little or not at all, the long-term trend is unequivocal: sea level has risen by about 25 cm from the 1880–1900 average, and it has been **rising at about 3 mm per year** over the 1993–2025 period.

Close examination of data also shows that the annual rate of rise has increased – from 2.0 mm a year in 1993 to 4.4 mm a year in 2023.[28]

What causes sea level rise?

There are two main reasons for sea level rise and both are connected to climate change. First, warming causes melting of land-based ice sheets and glaciers which results in the addition of fresh water to the oceans.

Ice loss, particularly in Greenland and Antarctica, has been the largest contributor to the sea level rise over the last couple of decades.

Secondly, ocean temperatures are rising because over 90% of heat accumulating on Earth due to the strengthened greenhouse effect (more on it on page *21*) is absorbed by the oceans. Warming oceans lead to **water expansion** which has contributed to about one third of sea level rise over the last decades.

The rise is uneven over the world

The averaged global sea level rise masks very different patterns in different places over the planet. Sea levels rise significantly in some parts of the globe, remain relatively flat in others, and even decrease in some areas.

In some coastal cities, particularly where land is also subsiding (e.g., Galveston on the Gulf Coast in USA, or Bangkok in Thailand) local sea level rise is **considerably higher** than the global average.[29]

What does the future hold?

Sea levels will continue rising for centuries to millennia because of melting ice even if warming were to stabilise. IPCC projects global mean sea level to rise between 0.28 and 1.01 metres in 2100, depending on future greenhouse gas emissions.[30] Some regions will see higher levels than the global average.

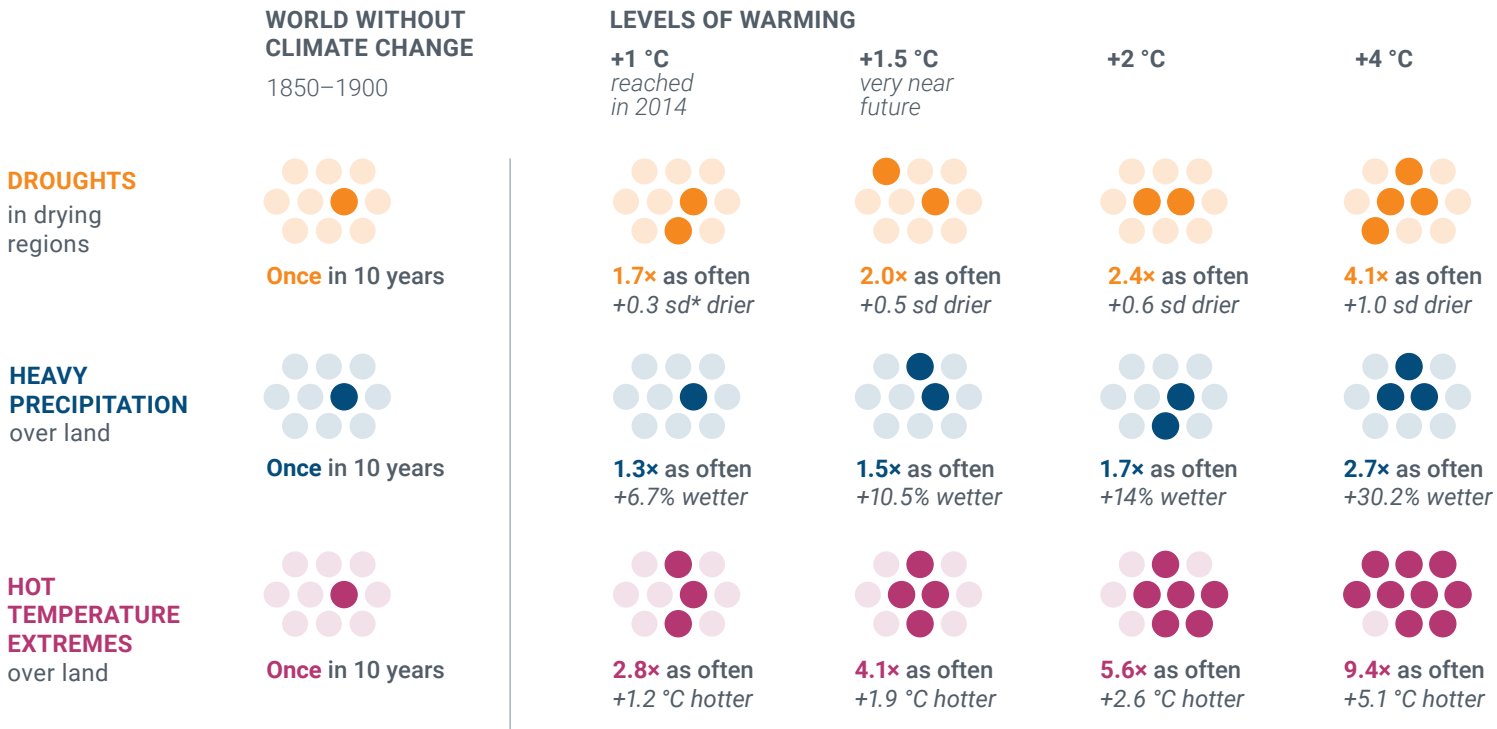
By 2300, IPCC projects up to 3 metres of global sea level rise even for the scenario of keeping global warming below 2 °C – a target that current emissions trajectories make increasingly unlikely. Scientists also warn that higher sea level rise cannot be ruled out due to deep uncertainties in ice sheet processes (i.e. how fast ice sheets may disintegrate and collapse).

Regardless, **even a relatively small change** in sea level can have **damaging impacts in coastal areas** due to much worse storm surges, flooding, salinisation of freshwater sources, erosion of land, and infrastructure damage. Future sea level rise poses a serious threat to hundreds of millions of people living around the coast.

Some governments in affected areas are trying to prepare for these future risks. For example, the Netherlands, where one third of the land already lies below sea level, is **investing in infrastructure to protect** the country from a sea level rise of up to 1 metre by 2050 and already planning for a rise of up to 5 metres.[31] However, protection is costly, and people in many parts of the world cannot afford such expenses.

Extreme Weather Is Growing More Severe

Extreme weather events will occur more frequently and with greater intensity with any rise in global temperature.



* For agricultural and ecological drought a fraction of standard deviation (sd) shows how much annual soil moisture shifts away from its historical average for a specific area.

Fakta o klimatu, CC BY 4.0

Data source: IPCC, AR6 WGI SPM

Extremes such as heat waves, droughts, and heavy rainfall occur more frequently and with greater intensity in a world under climate change. With any additional warming, extremes become greater and more pronounced.

The graphics depict **projected changes in frequency and intensity of selected extreme weather events**. Each dot represents one year in a hypothetical set of ten years. Dark dots indicate years with extreme events occurring, light dots years without the extremes. Events that occurred once per decade in the pre-industrial period now occur with higher frequency and are also more intense.[32]

Where are we now on the graph? Warming is now (as of 2026) about 1.3 °C relative to the pre-industrial era (see *How Much Is the Planet Warming? on p. 13*) and the 1.5 °C threshold is expected to be reached around 2030.

Weather extremes are connected

Many extreme events are linked to one another in a **complex web of mutual influences**. For example, heavy precipitation can trigger floods, hot temperature extremes exacerbate or prolong droughts, and droughts may then create conditions for wildfires.

Global warming leads to **more droughts and heavy precipitation at the same time**. Rising temperatures boost evaporation, which can cause more droughts and also bring more moisture to the atmosphere. A **warmer atmosphere can contain significantly more moisture** (about 7% more water vapour for each 1 °C), which leads to more intense storms, rainfall, and tropical cyclones.

With climate change, there is slightly more precipitation globally, but the **spatial and temporal distribution of this precipitation changes significantly**. In general, wetter areas are getting even wetter (tropical and polar regions), whereas drier areas are getting even drier (subtropics like the Mediterranean, the US Southwest, and parts of southern Africa). We are also seeing a shift toward fewer but more intense precipitation events: there are fewer days of rain and instead longer periods of droughts interrupted by downpours.[33]

Are all extreme weather events caused by climate change?

Not necessarily. Strictly speaking, any given event is never caused by global warming, but the likelihood of it occurring increases with global warming. The role of climate change can be assessed for each event in what is called **attribution studies**. What do they tell us?

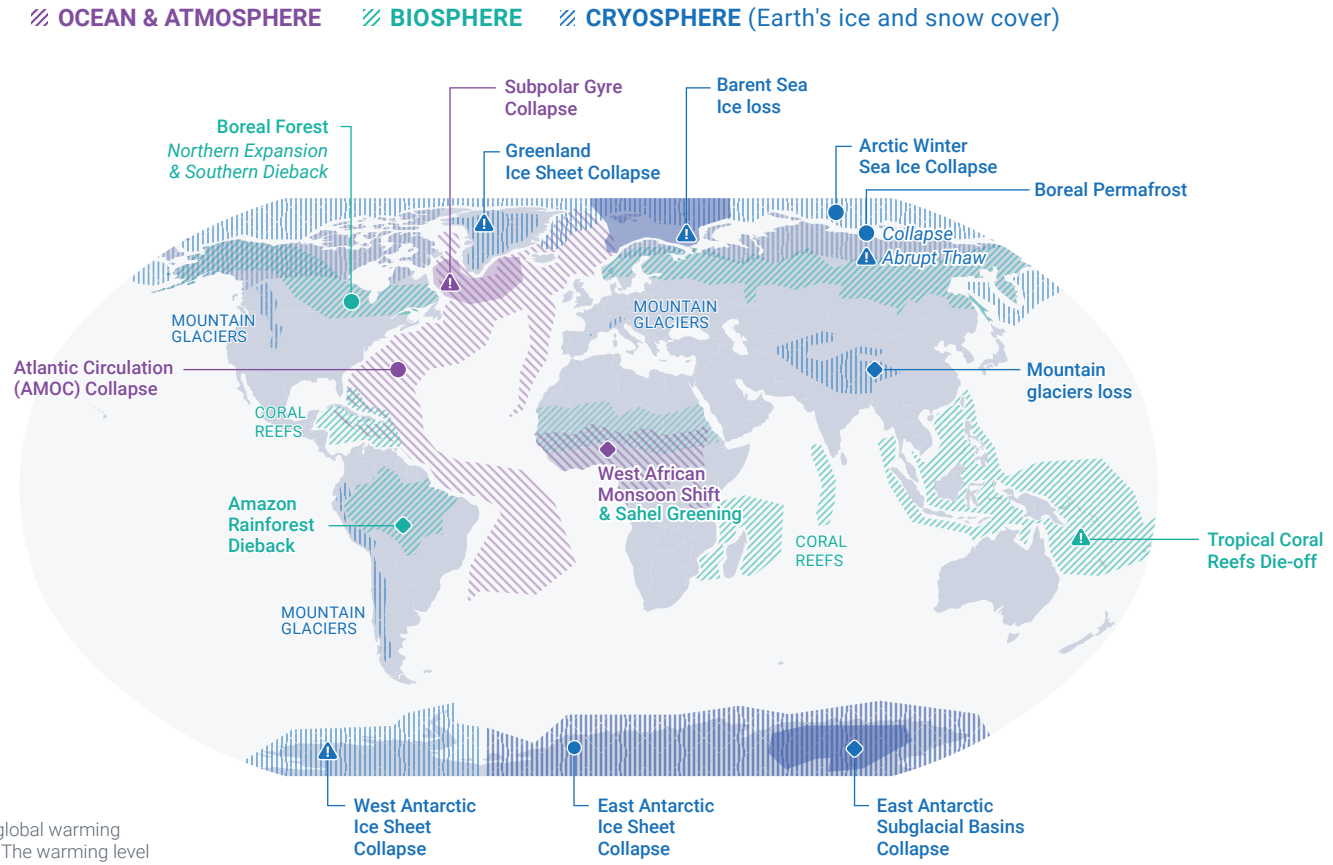
→ Heatwaves are very often much more likely to occur because of present warming. In a recent study published in Nature, a group of scientists concluded that out of 213 major heatwaves between 2000 and 2023, climate change has made 55 (over a quarter) of them 10,000+ times more likely – these heatwaves would have been virtually impossible without human influence.[34]

→ Some extremes are somewhat more likely to occur due to global warming. For example, the heavy precipitation, which brought devastating floods to Central Europe in September 2024 or to Spain in October 2024, was about twice as likely to occur because of climate change.[35]

→ For some events, however, the link to climate change remains weak or absent. A study of the severe drought in southern Madagascar in 2019–21, which caused significant food insecurity, concluded that climate change played only a limited role – natural rainfall variability was the primary driver.[36]

Map of Climate Tipping Points

Beyond certain critical thresholds, Earth's response to warming is no longer linear or predictable. Triggering these tipping points brings large-scale, irreversible changes to our planet.



The temperatures refer to global warming above pre-industrial levels. The warming level in 2025 is approx. 1.3 °C.

Fakta o klimatu, CC BY 4.0

Data source: McKay et al. (2022)

When warming exceeds tipping points, ecosystems, currents, and ice sheets can no longer maintain stability. While the threshold and the pace vary for each system, the resulting changes are irreversible. For humanity, this means moving beyond the stable climate that supported our civilization for millennia.

What are climate tipping points?

Imagine you are leaning over the side of a canoe to look at fish. You can lean quite far and the boat stays upright. But then at a specific point the centre of gravity shifts so much that the boat, carrying you with it, starts to inevitably flip over. No matter what happens next, the system's own physics finish the job. Similar dynamics work in ecosystems, ice sheets or ocean currents – warming beyond a certain level pushes them out of stability.

Major climate tipping points include:

Ice sheets collapse

There are a few major ice sheets on Earth and each is exposed to its own dynamics. Two of them, the Greenland and West Antarctica ice sheets, are already in the state of accelerated loss and may be at or near their tipping points.

The Greenland ice sheet covers 1.7 million km², roughly 80% of Greenland, and is over 3,000 m thick. As a mountain of ice, its surface lies in high, cold air. But with global warming, the surface melts and the ice sheet gets thinner. The **surface then drops into lower, warmer altitudes of the atmosphere, which causes even more melting**. Once the ice sheet drops below a certain critical elevation, it can no longer grow back even if we stop emissions, because the surface is now permanently in a warm-air zone. The complete melting of the Greenland ice sheet would cause a global sea level rise of 7 m and would probably take several thousand years.[37]

The West Antarctica ice sheet covers 2 million km² with ice over 2,000 m thick. Most of the **ice sits on bedrock that is below sea level and even slopes downward** as you go further inland. The warming ocean water melts the ice and causes it to retreat. Once the retreat reaches a downward slope in the bedrock, the loss of ice may become self-sustaining. This is because if the slopes go down toward the interior, it exposes a taller, thicker face of ice to the ocean, and thicker ice flows faster into the sea (more weight means more pressure pushing ice out to sea). The complete collapse of the West Antarctic ice sheet would raise global sea levels by up to 5 m in several thousand years.

Coral reefs die-off

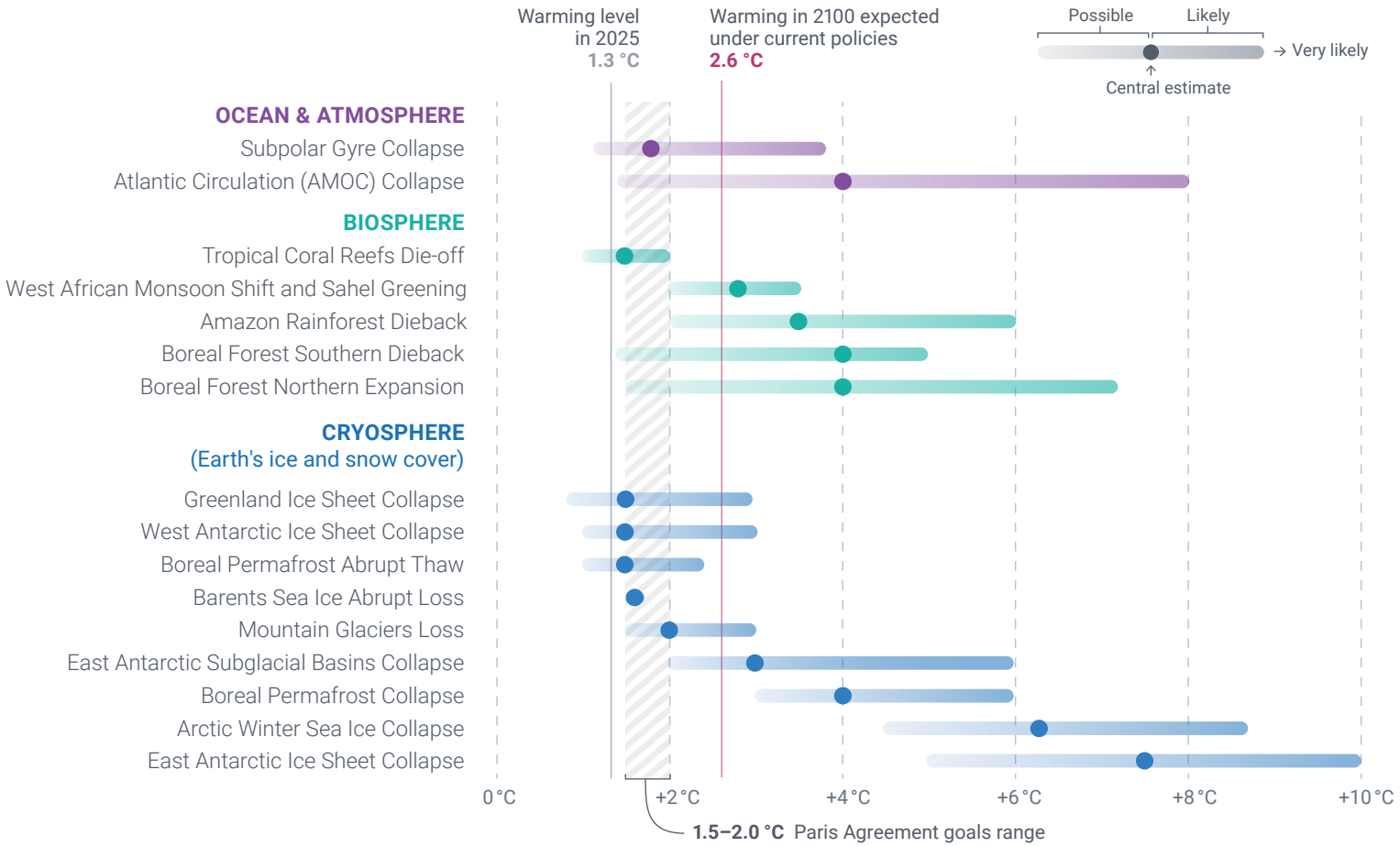
Coral reefs are built on a delicate partnership: the corals provide the limestone architecture, while microscopic algae living within them provide food via photosynthesis. When the sea becomes too warm, the algae begin producing harmful chemicals, forcing the coral to expel them. This leaves the coral white and starving – a process known as **bleaching** – and it dies if the water doesn't cool down quickly. These events are abrupt; a single marine heatwave can turn a vibrant reef ecosystem into a graveyard in just a few months.

Large parts of the Great Barrier Reef have recently experienced recurring bleaching events and many corals have died. Scientists estimate that **at 1.5 °C of global warming, 70–90% of tropical and subtropical reefs will be lost,** with near total loss by 2.0 °C. Coral collapse will eliminate one of the Earth's richest ecosystems in biodiversity, affecting a vast marine food web, and the livelihoods of hundreds of millions of people.

The text with other tipping points continues on p. *30*.

The Thresholds of Climate Tipping Risks

Five of Earth's vital climate components are already at risk of irreversible change. At 2 °C of warming, triggering seven tipping points becomes likely, while a further six move into the range of risk.



Fakta o klimatu, CC BY 4.0

Data source: McKay et al. (2022)

The risk of crossing climate tipping points is an imminent reality, not a distant concern. Within the Paris Agreement warming range of 1.5 °C to 2 °C, several systems are already likely to tip. These include the collapse of major ice sheets, the widespread die-off of coral reefs, and the abrupt thawing of permafrost.

Major climate tipping points include:
(continuation from p. *28*)

Permafrost thaw and collapse

Permanently frozen soils and sediments in vast boreal regions of Siberia and North America contain more carbon than is currently in the atmosphere – and as permafrost thaws, this carbon can be partially released as CO_2 and methane.

While gradual thawing is already underway, warming beyond 1.5 °C may trigger **abrupt permafrost thaw** in some regions, nearly doubling emissions compared to gradual melting.

Beyond 4 °C of warming, the abrupt drying of carbon-rich areas could lead to **permafrost collapse** or a **"compost bomb"** effect. In this scenario, the rapid decomposition of organic material produces its own heat, which further accelerates decomposition in a self-sustaining cycle. The resulting release of greenhouse gases could raise global temperatures by an additional 0.2–0.4 °C in just 50 years.[38]

Atlantic circulation collapse

The Atlantic Meridional Overturning Circulation (AMOC) acts as the planet's central heating system, carrying massive amounts of heat from the tropics toward the North Atlantic and warming Northwestern Europe. This circulation is self-sustaining: salty tropical water cools and becomes denser in the North Atlantic, causing it to sink and create deep currents flowing back south.

Global warming adds freshwater to the North Atlantic – from both the melting Greenland ice sheet and increased Arctic rainfall – which prevents the water from sinking. Studies suggest the **AMOC has already weakened by about 15%**. While a total collapse is considered unlikely during the 21st century, it becomes likely if warming reaches approximately 4 °C. Such a collapse would lead to regional cooling in the North Atlantic of 4–10 °C, profoundly altering the climate of Europe.

Closely linked to the AMOC is the **Labrador Sea Subpolar Gyre**, a smaller rotating current between Greenland, Canada, and Iceland. It is more vulnerable and may reach a tipping point much sooner – at roughly 1.8 °C of warming. Its collapse would cause regional cooling of 2–3 °C and increase weather extremes in Europe.

Amazon rainforest dieback

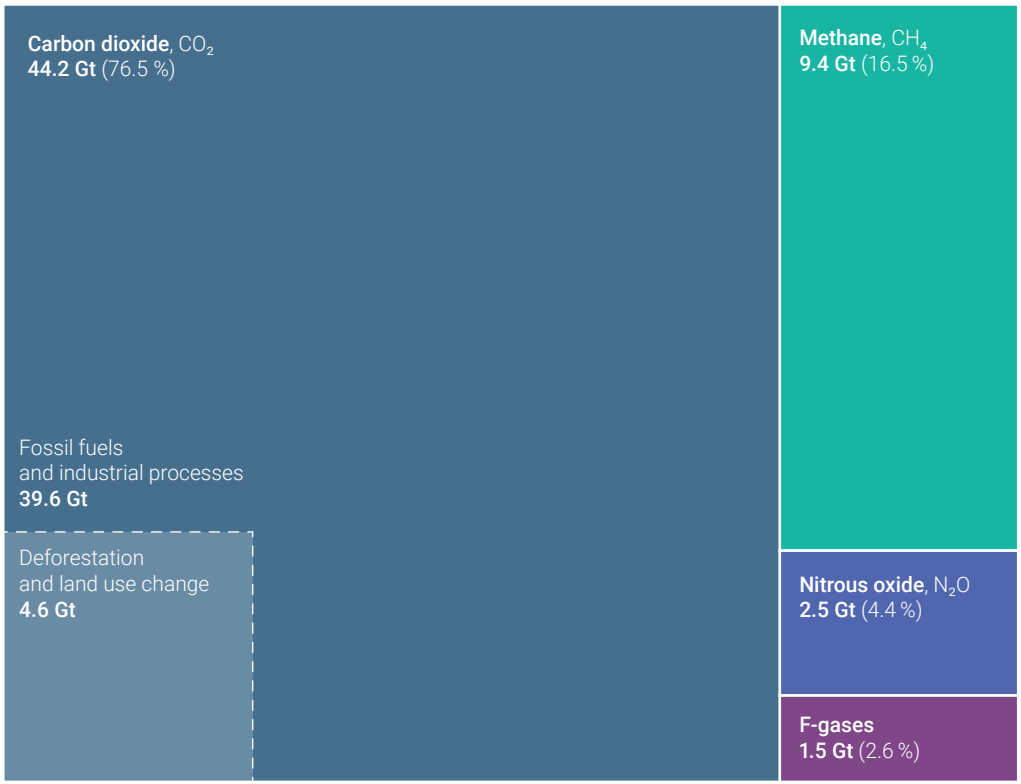
The Amazon, the largest and most biodiverse rainforest on Earth, was historically a vital carbon sink. However, a combination of deforestation and climate-induced droughts and wildfires (about 18% of the forest has been lost as of 2021[39]) has caused the **Amazon** basin as a whole to become **a net carbon source**.

As warming continues, the trend will only worsen – rainfall will decline and dry seasons will lengthen in southern and eastern areas of the forest. Scientists estimate that at 3.5 °C of warming, a **partial dieback affecting roughly 40% of the Amazon** becomes likely, shifting the region into a degraded, or savanna-like state. This self-reinforcing drying could be triggered even sooner if deforestation reaches 20–25%. The dieback would further disrupt regional rainfall and raise global temperatures by approximately 0.1 °C over the following century.

Which Greenhouse Gases Matter Most?

While CO_2 emissions are the dominant force behind global warming, methane (CH_4), nitrous oxide (N_2O), and F-gases also play a significant role in enhancing the greenhouse effect.

Global greenhouse gas emissions (in CO_2eq for 2024)



Emissions of all greenhouse gases are converted to CO_2 equivalents (CO_2eq) using their global warming potentials with a 100-year time-horizon (**GWP100**). This indicates how much CO_2 released to the atmosphere would have the same warming effect over a given period and it is the common way of reporting emissions of different gases in a single measure.

Fakta o klimatu, CC BY 4.0

Data source: EDGAR 2025, GCB 2025

Greenhouse gases differ in two vital ways: their potency and their permanence. Some, like methane, are powerful at trapping heat but vanish from the atmosphere within a decade. Others, like CO_2, have a lower impact per ton but linger for centuries or millennia, locking in warming for generations to come.

The graph shows annual global human-induced emissions of the main greenhouse gases for 2024.[40] To compare emissions of different gases, scientists convert them into **CO_2 equivalents (CO_2eq.)**. This is done using a metric called **Global Warming Potential (GWP)**, which calculates how much heat a gas traps over a 100-year period compared to CO_2. For example, since methane is roughly 28 times more potent than CO_2 over a century, one ton of methane is recorded as 28 tons of CO_2eq[41]. Using this scale, CO_2 alone is responsible for roughly three-quarters of the total warming impact of today's emissions.

What are the main anthropogenic greenhouse gases?

Carbon dioxide (CO_2) is the primary driver of climate change, released mainly through the burning of fossil fuels (coal, oil, and gas) and deforestation. A significant portion of CO_2 we emit today will remain in the atmosphere for centuries to millennia, gradually accumulating over time.

Methane (CH_4) is the second largest contributor to warming, arising primarily from agriculture (livestock and rice), leaky fossil fuel infrastructure, and waste decomposition. While it is about 80 times more powerful than CO_2 over a 20 year period, it breaks down in the atmosphere after only about a decade. This means that while methane causes intense short-term warming, reducing its emissions offers the fastest way to slow global warming in the near future.

Nitrous oxide (N_2O) is released mainly in agriculture due to excessive use of fertilisers rich in nitrogen. It is 273 times more powerful than CO_2 and it stays in the atmosphere for over a century. The rise of these emissions has been accelerating in recent years.

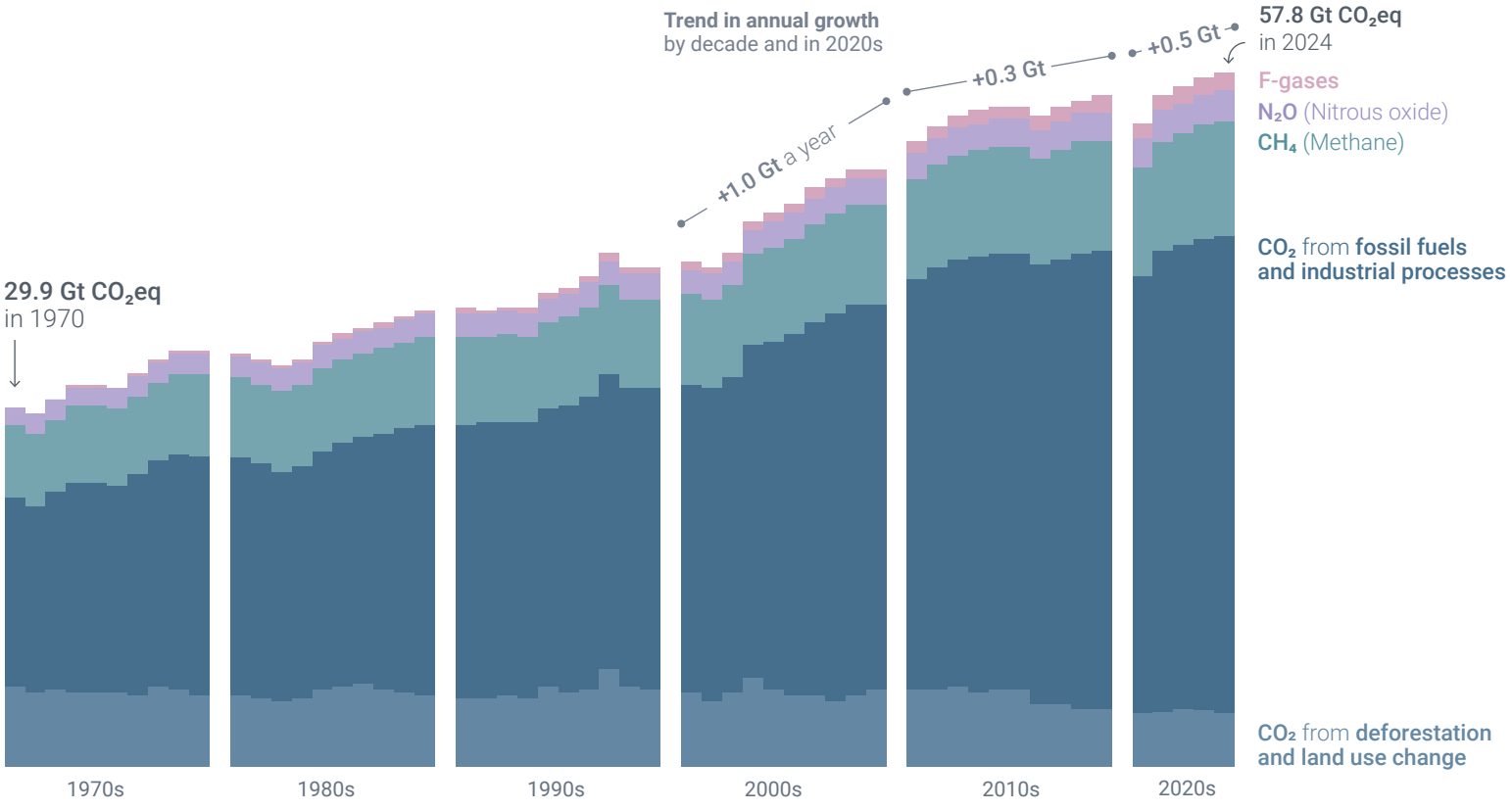
Fluorinated gases (F-gases) are man-made gases used mostly in refrigeration and air conditioning. While emitted in smaller quantities, they are "super-pollutants" – some are thousands of times more effective at trapping heat than CO_2. The emissions of F-gases have risen rapidly, but international agreements are already in place to phase down their use in favour of climate-friendly alternatives.[42]

A note on water vapour

It is often argued that water vapour is the most abundant greenhouse gas. While true, it is not a direct "driver" of warming caused by humans. Instead, water vapor acts as an **amplifier**: as CO_2 and other gases warm the air, the atmosphere holds more moisture, which then traps even more heat. This **feedback loop** is a well-understood part of climate modeling and is triggered by the primary human-induced gases described above.

Global Emissions Continue to Grow

Global greenhouse gas emissions have nearly doubled over the past 55 years. Their growth peaked during the 2000s and slowed significantly in the 2010s, but has picked up again in the 2020s.



Fakta o klimatu, CC BY 4.0

Data source: EDGAR 2025, GCB 2025

Global greenhouse gas emissions are still rising as of the end of 2025, albeit at a slower pace than two decades ago. To meet international climate targets and to avoid even more dangerous warming, global emissions need to fall substantially.

The graph shows the trend in human-induced greenhouse gas emissions since 1970.[43] Emissions of all gases are converted to **CO_2 equivalents (CO_2eq)** – the amount that would have the same warming effect as CO_2 over 100 years.

Emissions have increased steadily over the long term, with the sharpest rise between 2001 and 2012. Since 2010, the pace of growth for CO_2 from fossil fuel combustion and industrial processes, methane (CH_4), and nitrous oxide (N_2O) has averaged about 1% annually. F-gases have seen the steepest rise, exceeding 4% per year over most of the period, driven by growing global demand for air-conditioning and refrigeration. Only CO_2 emissions from land use and forestry (LULUCF) have remained stable or slightly declined.

Which gas is most important for warming?

Carbon dioxide (CO_2) is responsible for over half of human-induced global warming, but other anthropogenic gases – most notably methane (CH_4) – play a significant role. Methane is far more potent at trapping heat than CO_2, but it is short-lived, remaining in the atmosphere for only about a decade. Thus, methane influences the **rate** of warming in the short term, but because CO_2 accumulates in the atmosphere for centuries, the **cumulative CO_2 emissions** remain the primary driver of long-term temperature rise.

Research synthesized by the IPCC shows a near-linear relationship between CO_2 emissions and warming: **every 1,000 Gt CO_2 of cumulative emissions adds approximately 0.45 °C to global temperatures.** This allows us to calculate a "carbon budget" – the total amount of CO_2 humanity can emit before we cross specific thresholds, such as 1.5 °C or 2 °C (see *The Carbon Budget: How Much Remains? on p. 35*). To stop further warming, we must not only reduce annual emissions but eventually stop adding to this cumulative total entirely.

What influences the trend in future emissions?

Global emission trajectories are determined by the balance between economic ambition and technological transformation. In emerging economies, the primary driver of rising emissions is rapid **economic development (GDP per capita)** and urbanization. While population growth contributes, it is the massive expansion of infrastructure, energy use, and industrial output required to lift billions of people into the middle class that exerts the greatest upward pressure.

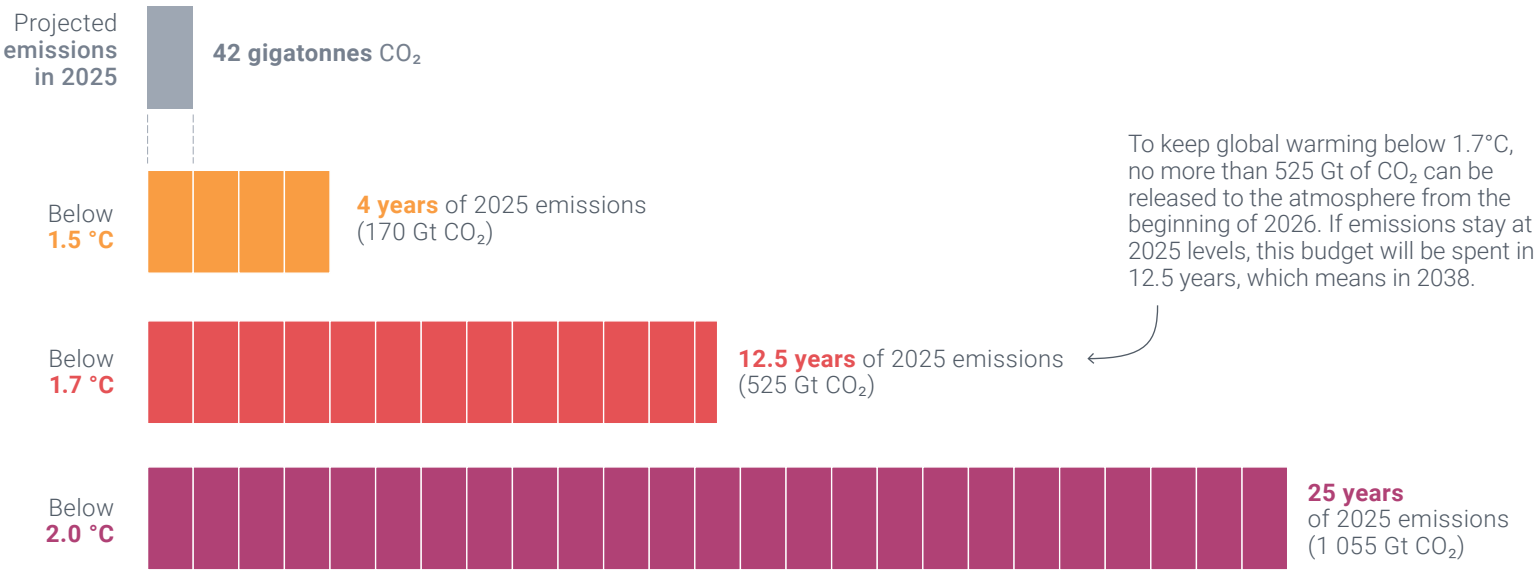
In contrast, many developed nations have begun to **decouple economic growth from emissions**. In these regions, improvements in **energy efficiency** and the **decarbonization of power grids** are actively displacing fossil fuels. The rapid scaling of clean technologies – such as wind, solar, and electric vehicles – is now a major downward force on emissions.

According to the International Energy Agency (IEA), if current policy momentum continues, these efficiency gains and clean energy shifts could cause global emissions to **peak by 2030**.[44] However, the speed of the subsequent decline depends on whether technological improvements can outpace the legitimate economic aspirations of the developing world.

The Carbon Budget: How Much Remains?

If global CO_2 emissions continue at 2025 levels, the remaining carbon budget for staying below 1.5 °C of global warming will be exhausted in 2030; for staying below 2 °C, it will be spent in 2051.

How much CO_2 can we still emit from 2026 onward
to keep global warming below a certain temperature?



Estimates are based on a 50% likelihood
of staying below the given temperature threshold.

Fakta o klimatu, CC BY 4.0

Data source: GCB 2025

The extent of global warming depends primarily on how much more CO_2 humans add to the atmosphere. Limiting warming to well below 2 °C, in line with the Paris Agreement, is becoming increasingly challenging.

The extent of global warming is dictated by a simple, near-linear rule: **for every 1,000 Gt CO_2** added to the atmosphere, **global temperatures rise by approximately 0.45 °C**.[45] Because CO_2 persists for centuries, its warming effect is cumulative. This allows scientists to calculate a **"carbon budget"** – the maximum amount of CO_2 we can still emit to limit warming to a specific level.

The graph shows the remaining carbon budgets from the start of 2026 **to limit global warming to 1.5 °C, 1.7 °C, and 2 °C**.[46] To stay within these limits, the world must eventually reach **Net Zero**, the point where we stop adding more CO_2 to the atmosphere than we remove. If global emissions continue at 2025 levels, the remaining budget for 1.5 °C will be exhausted by 2030, and the budget for 2 °C will be spent by 2051.

These budgets offer a **50% likelihood** of success. For example, if the world emits approximately 1,055 Gt CO_2 from 2026 onwards, there is an even chance that warming stays below 2 °C. If we require a higher degree of certainty, the "allowable" budget becomes even smaller.

While CO_2 is the primary driver, these estimates also account for the cooling effects of aerosols and the warming effects of other gases like methane.

What would it take to stay below 1.5 °C, 1.7 °C, and 2.0 °C?

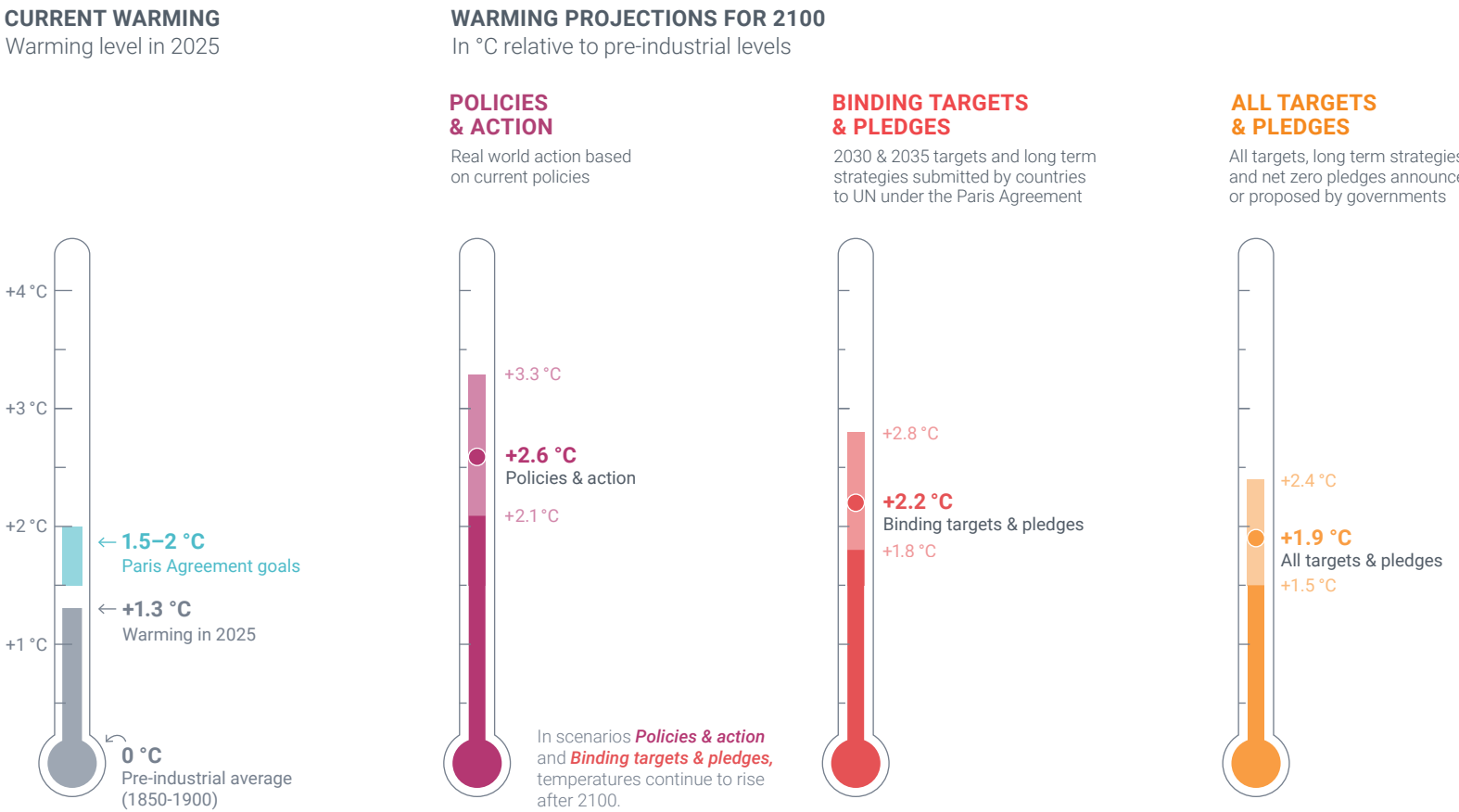
The Minimum Warming Path (1.5 °C): The budget for 1.5 °C is now extremely tight: **only 170 Gt CO_2**. To stay within it, the world would need to cut emissions by about 11% (4.6 Gt) every single year, reaching Net Zero by 2034. As global emissions are not expected to drop substantially by 2030, this path is increasingly viewed as unrealistic. While theoretically possible, such scenarios now rely on temperatures "overshooting" 1.5 °C and being brought back down later through large-scale CO_2 removal.

The Net Zero 2050 Path (1.7 °C): If global emissions begin to drop in 2026 and reach Net Zero by 2050 – as many countries have pledged – emissions would need to decrease by roughly 4% per year. This rate of decline is comparable to the unprecedented drop seen in 2020 during the COVID-19 pandemic. On this trajectory, the world would release just over 500 Gt CO_2, aligning with the **1.7 °C budget**. This remains an extremely ambitious goal, especially as many major emitters have set Net Zero targets for 2060 or 2070.

The Plateau and Decline Path (2 °C): Imagine a scenario where emissions plateau at 2025 levels until 2030, then gradually decrease to Net Zero by 2070. This would require an annual reduction of 2.5%. Under this plan, the world would emit roughly 1,050 Gt CO_2 – significantly more than the previous scenarios, but still within the **2 °C budget**. While this path is more in line with current national pledges, it is far from guaranteed; most countries are not yet on track to meet their existing promises (see more in *Projections of Warming in 2100 on p. 37*).

Projections of Warming in 2100

With current policies, the world is on track to reach approximately 2.6 °C of warming by the end of this century. Only if all climate pledges are met will the world limit warming to about 1.9 °C in 2100.



Fakta o klimatu, CC BY 4.0

Data source: CAT Thermometer, November 2025 © 2025 by Climate Analytics and NewClimate Institute.

Future temperatures depend on two factors: future greenhouse gas emissions and the Earth's physical response to those gases. Although significantly less warming is now expected by 2100 than was projected just a decade ago, current policies and climate pledges remain insufficient to limit warming to well below 2 °C – the central goal of the 2015 Paris Agreement.

There are two main sources of uncertainty when scientists try to estimate the levels of future warming. One is **uncertainty about future greenhouse gas emissions**, which directly affect warming. The other is the **uncertainty in the climate sensitivity and the precise response of the climate system to higher greenhouse gas concentrations** and other changes in the atmosphere. Both are shown in the graph.

The three depicted scenarios represent different emission pathways.[47] For each scenario, scientists estimate the amount of greenhouse gas emissions expected to be released in the future.

With these estimates of future emissions, scientists run a climate model to project future warming. But even with their increasing understanding of the climate system, they do not know precisely how the Earth will respond – for example, exactly how much CO_2 the oceans will absorb or how clouds will change as the planet warms.

Because of these factors, warming projections are always given as a **likely range**. For example, in the Policies & action scenario, the projected range for 2100 is 2.1 °C to 3.3 °C. The **best estimate** of 2.6 °C represents the median outcome (a 50% chance of being higher or lower). The full range indicates where warming is likely to fall with a 68% level of confidence.

What are the three scenarios?

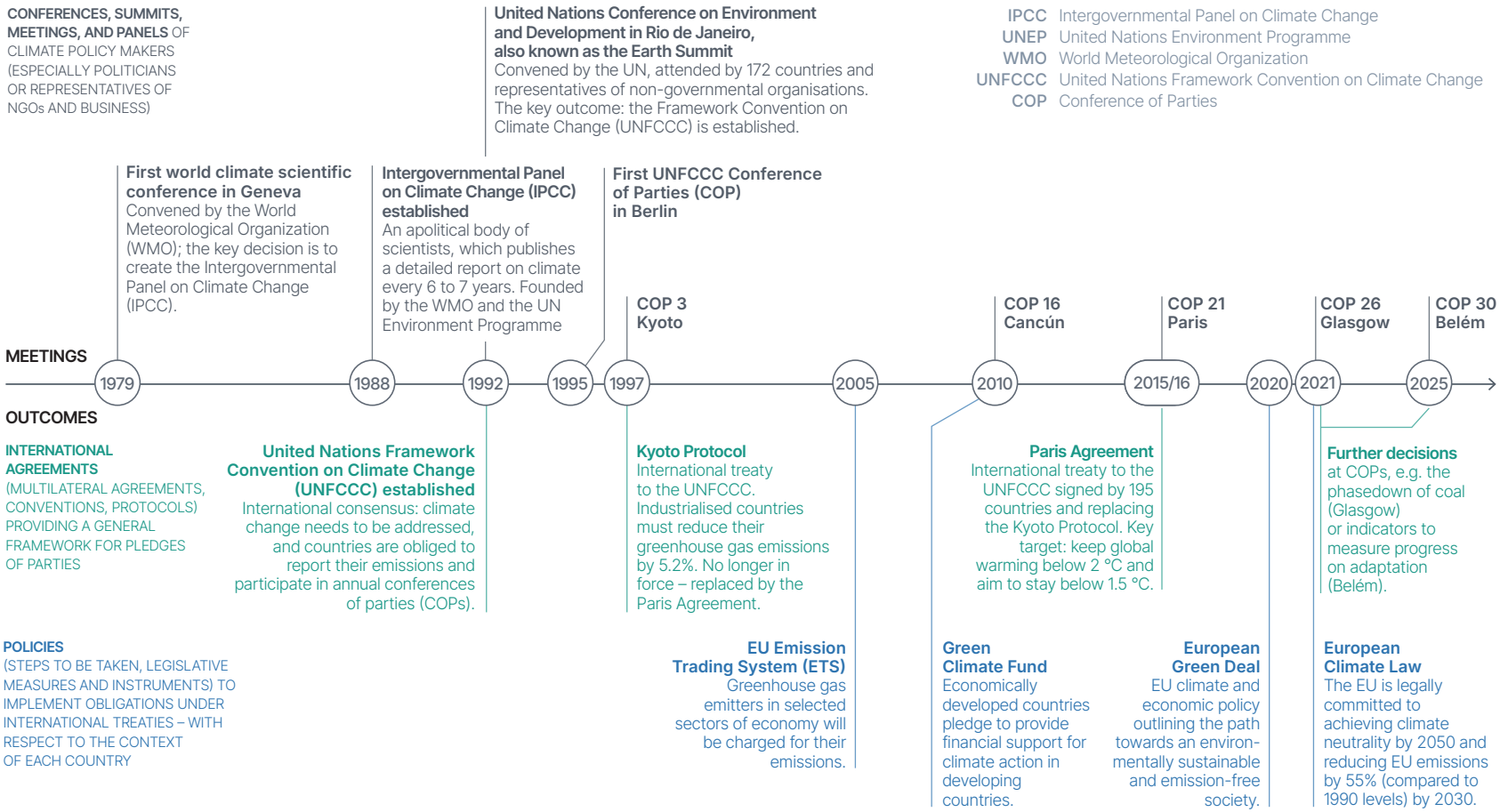
Policies & action: This scenario is based on current actions and policies already enacted into law or under significant development. It is important to note that this is not a "worst-case" scenario; it actually assumes that current climate policies enacted into legislation (such as that in the EU) will be successfully implemented and emissions reduced. If these measures are rolled back or delayed, warming could be significantly higher than this scenario projects.

Binding targets & pledges: This scenario assumes countries meet the formal emission reduction targets and long-term strategies they have submitted under the UN Paris Agreement. This covers both the 2030 & 2035 targets and net zero emissions pledges, including pledges for 2050 (EU, UK, Japan, South Korea, Canada, Australia, Brazil, and other countries), 2060 (China, Indonesia), and 2070 (India).

All targets & pledges: This is the most optimistic pathway. It assumes that governments will meet every climate target, proposal, and informal announcement they have made, even those not yet formally submitted as binding international pledges. This is currently the only scenario in which warming is projected to peak before 2100 and stabilize rather than continue rising.

The History of International Climate Agreements

Key milestones in global effort to protect the world's climate – important summits, panels, negotiations, and treaties.



Fakta o klimatu, CC BY 4.0

Data source: UN, IPCC, European Commission

Climate change is a global challenge requiring long-term collaboration on both national and international levels. While international negotiations can be slow and cumbersome, they establish an important framework for global climate protection and emissions reduction.

The timeline in the infographics highlights the most important global and European events in climate policy. It begins with the First World Climate Conference in 1979 and extends to the European Climate Law and annual Conferences of Parties (COPs), where world leaders meet to negotiate further progress following the Paris Agreement.

What is IPCC and how does it work?

The **Intergovernmental Panel on Climate Change (IPCC)** is a body of experts from 195 countries who review the latest research and publish comprehensive **Assessment Reports** roughly every six to seven years. Rather than conducting new research, the IPCC synthesizes the most reliable scientific findings to provide a robust evidence base for policymakers. These reports describe current trends and provide critical data on mitigation (reducing emissions) and adaptation (adjusting to change). Most modern climate strategies, including those of the EU, are informed by the IPCC's conclusions.

What is the Paris Agreement and why is it important?

Adopted in 2015, the Paris Agreement has achieved near-universal adoption. While almost every nation has signed and ratified the treaty (the only exceptions are Iran, Libya and Yemen), the United States occupies a unique position, having withdrawn in 2020, rejoined in 2021, and withdrawn again in early 2026.

The long-term goal of the agreement is to **keep the rise in global average temperature to well below 2 °C above pre-industrial levels** and pursue efforts to limit the increase to 1.5 °C (see *How Much Is the Planet Warming? on p. 13*). While many experts now consider the 1.5 °C target out of reach, the 2 °C target can still be achieved. Every tenth of a degree of warming prevented is vital for climatic stability, the health of ecosystems, and for the livelihoods of those who depend on them (see more on the climate tipping points on the pp. *27* and *29*).

The agreement is a **legally binding international treaty**. Under its framework, each country is required to develop its own climate action plan with specific targets for emissions reduction and adaptation. These plans must be updated every five years with increasing ambition, aiming for global **Net Zero** emissions in the second half of this century.

When the first round of targets was submitted to the UN in 2015, it was estimated that if all were fully implemented, the temperature increase would be only limited to about 3-3.5 °C above pre-industrial levels by 2100. As of the end of 2025, if all updated targets were implemented, the projected temperature increase dropped to about 2.3-2.5 °C according to UNEP.[48] Over the ten years since the Paris Agreement, projected warming has fallen significantly – but both ambition and implementation still fall short of its ultimate goals. (See more on the future warming estimates on the p. *37*).

Climate Solutions Across Sectors

Stopping greenhouse gas emissions requires distinct strategies for different sectors. In most cases, deep decarbonization relies on a combination of several solutions rather than a single fix.

A Energy efficiency B Electrification C Clean energy production D Changes in fuels and technologies E Carbon management and removal F Changes in consumer behaviour

WASTE MANAGEMENT (3 %)

D F Organic waste recovery (composting and biogas)

BUILDINGS (7 %)

B F Electrification of heating (heat pumps)

A F Building insulation and energy efficiency

AGRICULTURE (12 %)

F Shift to plant-based diets

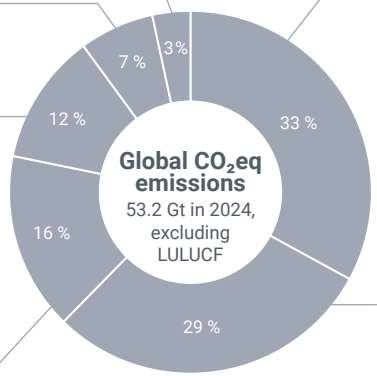
D E Regenerative and precision farming

TRANSPORTATION (16 %)

A B F Electrification of road transport

A F Public transport and active mobility

D Low-emission fuels for aviation and shipping



INDUSTRY (33 % of global emissions)

A B Electrification and energy efficiency

D Technological changes (steel, cement, and fertilisers production)

E Carbon capture and storage (CCS)

ELECTRICITY & HEAT PRODUCTION (29 %)

C Clean energy production

Emissions from forestry and land use (LULUCF: +4.6 Gt CO_2) are not shown in the chart.

DEFORESTATION & LAND USE CHANGE

E Sustainable forestry and land conservation

Fakta o klimatu, CC BY 4.0

Data source: EDGAR 2025; GCB 2025

Paths towards decarbonising various sectors of the economy fall into six distinct categories, coded A–F in the graphic.

Decarbonization *refers to the process of reducing or eliminating greenhouse gas emissions from the economy [49]. The ultimate goal is Net Zero – a state where any remaining emissions are balanced by an equivalent amount of carbon removal.*

A Energy efficiency

Reducing the amount of energy required to achieve the same result. Examples include thermal insulation for buildings, or waste heat recovery in factories. Electrification also drives efficiency, as electric motors waste far less energy as heat than internal combustion engines.

B Electrification

Replacing fossil-fuel-burning equipment with electric alternatives. Most road transport can switch to electric vehicles (EVs), while heat pumps can replace gas boilers in homes and industry. This strategy is most effective when the electricity comes from clean sources.

C Clean energy production

The global power sector is undergoing a massive transformation. In 2004, **wind and solar** provided less than 1% of global electricity; today, they provide nearly 20%. When combined with nuclear, hydro, and biomass, low-emission sources now account for **over 40% of global generation**.[50] This shift is making it possible to gradually phase out coal and gas power plants or use them only as back-up sources. Reaching 100% requires expanding energy storage (batteries), strengthening transmission grids, and increasing demand flexibility.

D Changes in fuels and technologies

In sectors where electricity isn't practical (like long-haul shipping or aviation), a **transition to low-emission fuels** like green hydrogen, methanol, ammonia, or advanced biofuels are an option. **Innovations in production processes** include modifications to cement compositions, the use of hydrogen in iron production, or regenerative and precision farming in agriculture. **Reducing methane emissions** – by fixing leaks in fossil fuel infrastructure, improving livestock management, and capturing gas from decomposing organic waste – remains one of the highest-impact near-term measures given methane's potency as a greenhouse gas.

E Carbon management and removal

Mitigation: Carbon Capture and Storage (CCS) involves trapping CO_2 at the source (like a factory chimney) before it reaches the atmosphere, and pumping it underground.

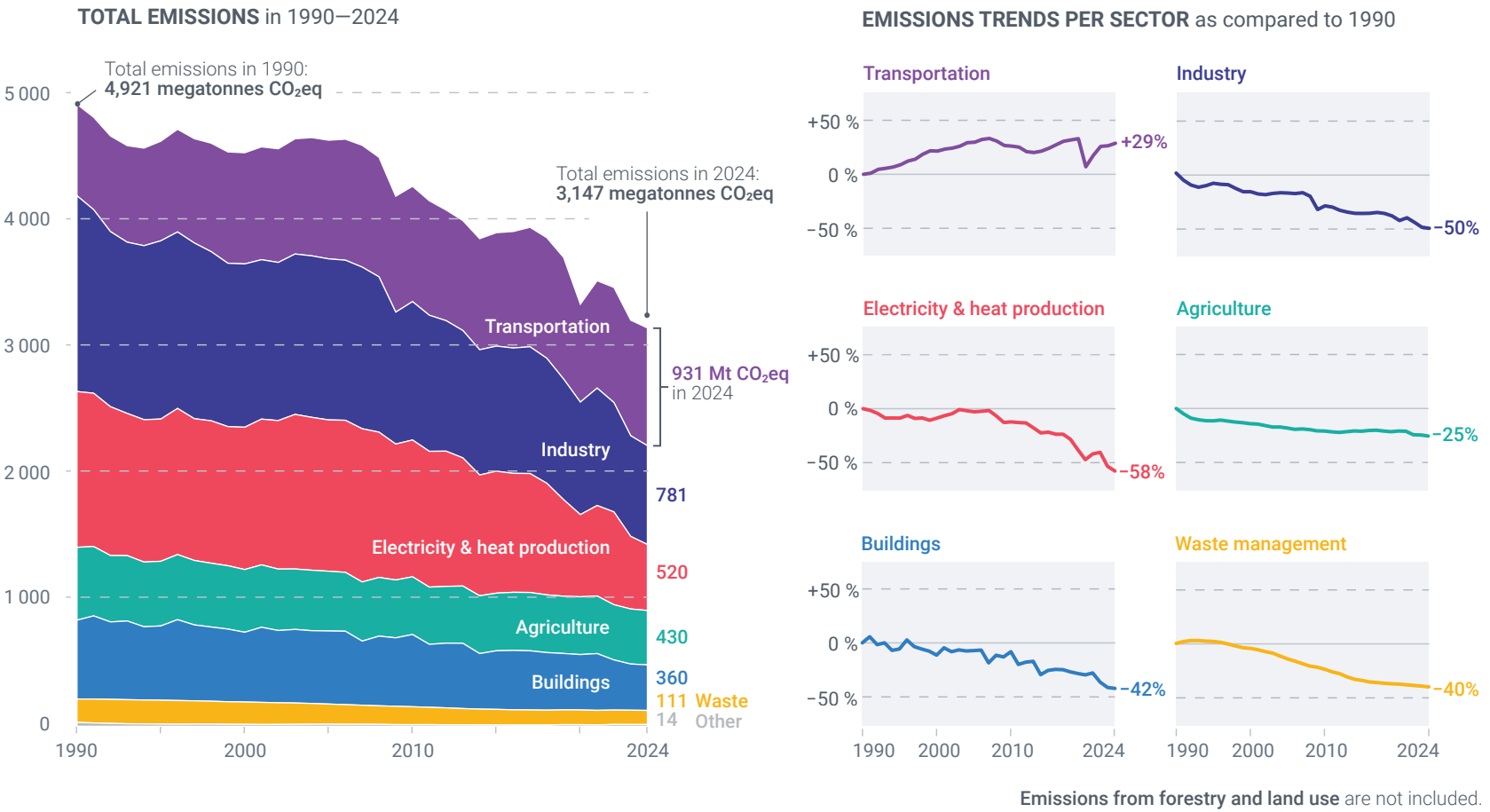
Carbon dioxide removal (CDR): This involves taking CO_2 out of the atmosphere. Natural methods include reforestation and soil restoration. Technological methods include Direct Air Capture (DAC), enhancing ocean alkalinity or biochar production, though most of these technologies remain expensive to scale.

F Changes in consumer behaviour

Individual choices can significantly reduce emissions. The highest-impact shifts include using public transit, adopting electric vehicles or opting for active travel (walking/cycling), reducing air travel, adopting plant-based diets, insulating homes or installing residential solar rooftops. These changes often provide co-benefits, such as improved public health and reduced urban congestion.

The Steady Decline of EU Emissions

Between 1990 and 2024 the EU emissions dropped by 36%. Emissions decreased in all sectors, except for transportation.



Emissions from forestry and land use are not included.

Fakta o klimatu, CC BY 4.0

Data source: Eurostat, env-air-gge

The European Union reduced its emissions by over one third between 1990 and 2024. This progress reflects a fundamental shift in how the region generates energy and manages its economy.

The emissions depicted include CO_2 from fossil fuel combustion and industrial processes, methane (CH_4), nitrous oxide (N_2O), and other greenhouse gases converted to **CO_2 equivalent** (CO_2eq). This measure indicates the amount of CO_2 that would have the same greenhouse effect over a 100-year period (See more on greenhouse gases on p. *31*).

The left side of the graph shows the total annual emissions for the EU-27, while the right side highlights relative changes by sector. In more than three decades, the total fell from 4.9 billion tonnes of CO_2eq to 3.1 billion tonnes,[51] with the pace of reduction accelerating over the last fifteen years. The sharp dip in 2020 reflects the COVID-19 economic slowdown, while the subsequent trend shows a return to policy-driven declines.

For comparison, 3.1 billion tonnes of CO_2eq in 2024 equals **7 t per capita**. This is above the global average (roughly 6.5 t per capita) and much higher than India (approx. 3 t), but significantly lower than China (approx. 11 t) or the USA (over 17 t).[52]

How did emissions change in each sector?

Values indicate per capita CO_2eq emissions in 2024; the change is between 1990 and 2024.

Transportation (2.07 t per capita; +29%): Passenger cars still produce about half of all transport emissions, with trucks and buses accounting for most of the rest. Emission growth was driven by a massive rise in the volume of passenger cars, trucks, and aviation. However, transportation emissions have broadly plateaued since 2007; while aviation continued to grow until 2019, road transport emissions have begun to stabilise.

Industry (1.73 t per capita; −50%): Most industrial emissions come from burning fossil fuels to create the high temperatures needed for producing steel, cement, and other materials; others result from internal chemical reactions. While heavy industries like steel, chemicals, and cement became more efficient, part of the decline reflects a shift away from energy-intensive manufacturing and the relocation of some production outside the EU.

Electricity and Heat Production (1.16 t per capita; -58%): The emissions result from burning fossil fuels to power the grid and district heating systems. Reductions have accelerated recently due to a rapid coal phase-out and the rise of competitive wind and solar power. The EU Emissions Trading System (ETS), which prices carbon, is a primary driver.

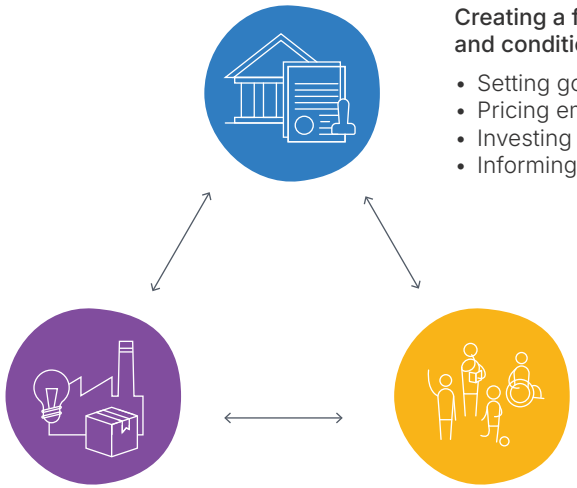
Agriculture (0.96 t per capita; −25%): Agricultural emissions are driven by three main sources: livestock methane from digestion and manure, soil nitrous oxide from fertilisers, and carbon dioxide from diesel-powered machinery. Emissions fell largely due to efficiency gains in livestock farming: the number of cows in the EU has decreased even as milk production has risen. Additionally, a shift from beef toward poultry has lowered meat's average carbon footprint.

Buildings (0.80 t per capita; −42%): Emissions mainly reflect the direct consumption of gas for heating and hot water in homes and offices. Over thirty years, building standards shifted from minimal energy requirements in some countries to EU-mandated "nearly zero-energy" new buildings. While some older buildings were renovated with better insulation, a large portion still requires upgrading.

Waste management (0.25 t per capita; -40%): Stricter EU standards succeeded in diverting much biodegradable waste from landfills to prevent methane formation. Recycling and composting, once niche in 1990, handled nearly 50% of municipal waste by 2024.

Who Drives Change? State, Business, and People

Decarbonization involves three main groups of actors: the state, businesses, and people. Success depends on their cooperation, as each holds different tools to reduce emissions.



The state

Creating a favourable environment and conditions for change

- Setting goals, establishing rules, standards, and regulations
- Pricing emissions, subsidizing low-emission solutions
- Investing in infrastructure
- Informing the public and raising awareness

Businesses

Decarbonising production and services

- Electrification and transition to clean electricity
- Energy and resources efficiency
- Implementing low-emission technologies
- Investment in innovations

People

Supporting decarbonisation and reducing emissions

- Active involvement at the state, municipal, and corporate levels
- Renovation of homes and other steps leading to energy savings and decarbonization of households
- Willingness to change consumer behavior where possible

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Out of the three key actors, the state holds a unique position. By setting the rules of the game, it creates the systemic framework that makes it easier for businesses and individuals to take the necessary steps toward a low-carbon future.

The State

The role of government – from local municipalities to supranational bodies such as the EU – is to create conditions for a transition that is swift, efficient, socially just, economically beneficial, and publicly supported. To achieve this, governments use a mix of four policy types:

Regulatory: Setting binding targets, performance standards (e.g., fuel efficiency for cars), and building codes.

Economic: Carbon pricing (taxes or trading systems), subsidies for clean technology, and tax reliefs for green investments.

Infrastructure Investment: Funding large-scale projects like modernized electricity grids, high-speed rail, and public charging networks that private investors may not fund alone.

Awareness & Transparency: Establishing rules for energy labeling, carbon footprint disclosure, and public awareness campaigns to help citizens make informed choices.

The scale of investment needed for the transition far exceeds what governments alone can provide. Therefore, one of the most vital functions of climate policy is to provide the **stability** and **clarity** needed to mobilize massive amounts of private investment.

Businesses

Companies are the engines of the technical transition. They decarbonize by:

→ Switching to clean electricity and electrifying manufacturing processes.

→ Improving resource efficiency and reducing waste through circular economy models.

→ Developing and scaling clean technologies and low-emission innovations.

→ Investing in the necessary changes.

While service-sector companies may only need to switch to clean electricity and operate from a low-energy building, heavy industries like steel, cement, and chemicals require a fundamental transformation of their production processes. These sectors are the most critical for achieving deep emission cuts.

People

The largest share of global emissions is linked to heavy industry and energy production, where individuals have limited direct control. However, **the active participation** of individual people – as citizens, voters, employees, and purchasers of goods/services – will be essential to the success of decarbonisation even in those sectors. All people can initiate, advocate for, and implement change.

Furthermore, people have direct influence over emissions from their homes, food, and transport. **Shifts in consumer behaviour** like using public transit or adopting a more plant-based diet and **investments** in insulation, heat pumps, rooftop solar, and electric vehicles provide immediate and significant emission reductions.

Other actors

The transition is also supported by international organizations (like the UN), non-governmental organizations (NGOs), trade unions, banks, and educational institutions, all of which help bridge the gaps between the state, businesses, and the public.

Endnotes

1 See Chapter 7 of IPCC AR6 WGI report. Intergovernmental Panel on Climate Change. (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to AR6*. DOI: 10.1017/9781009157896.

2 See D.1.1 in IPCC's AR6 WGI Summary for Policymakers: Intergovernmental Panel on Climate Change. (2021). Summary for Policymakers. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to AR6*. DOI: 10.1017/9781009157896.001.

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10 Copernicus Climate Change Service set the global temperature anomaly in 2024 relative to pre-industrial period at 1.6 °C. Copernicus Climate Change Service (2025). *Global Climate Highlights 2024*. [https://climate.copernicus.eu/global-climate-highlights-2024]

11 Temperature data come from NASA Goddard Institute for Space Studies, The GISS Surface Temperature Analysis version 4 (GISTEMP v4) [https://data.giss.nasa.gov/gistemp/]. Dataset accessed [2026-01-22].

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13 The value of 3.5 °C comes as the upper range of warming in the 2080-2100 period in IPCC's SSP2-4.5 scenario. This "Middle of the Road" scenario supposes slow progress toward sustainability with greenhouse gas emissions peaking between 2040 and 2050 and then steadily declining. It is by no means the worst case scenario – if emissions continue to rise throughout the 21st century, the warming might be even bigger. See Cross-Section Box TS.1, Table 1 in Intergovernmental Panel on Climate Change. (2021). Technical Summary. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to AR6*. DOI: 10.1017/9781009157896.002.

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23 The graphics including the numbers are based on Figure 7.2 of IPCC AR6 WGI report. Intergovernmental Panel on Climate Change. (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to AR6*. DOI: 10.1017/9781009157896.

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47 The graph is based on the Climate Action Tracker (CAT) Thermometer, version from November 2025, data were retrieved on 2026-02-04. Credit to: © 2025 by Climate Analytics and NewClimate Institute [https://climateactiontracker.org/global/cat-thermometer/].

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